

Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation

Abstract

Graph recommenders propagate preferences along walks of a logged graph whose edges exist only where an item was exposed. Debiasing re-weights edges by inverse propensities or completes them with doubly robust (DR) estimates. For multi-hop propagation this is biased: a walk that traverses an edge several times multiplies its estimate with itself. We give the exact bias and remove it with DRUP (Doubly Robust, walk-Unbiased Propagation), which replaces each repeated edge by its idempotent counterpart; Möbius inversion extends the correction to any odd number of hops. The guarantees require nuisances that do not depend on the log. In simulation, cross-fitting suffices only if the degree weights are fixed a priori; with degrees taken from the log, as in our experiments, the correction no longer reduces bias. We find no detectable change in accuracy: on Coat, Yahoo!R3, KuaiRand and KuaiRec the corrected and uncorrected DR operators are equivalent within 0.005 nDCG, and the most accurate method differs by dataset. The correction matters for score values, but only under oracle conditions: on a semi-synthetic KuaiRec log it removes an 18–29% overestimate of an allocation’s utility when propensities are true and weights fixed, while with estimated propensities all operators overstate it by 100–540%. The sample split that

satisfies the assumptions costs up to 0.06 nDCG on sparse logs. A real-data thinning audit of exposure invariance has limited power, and the selected configurations are not invariant on Yahoo!R3 and KuaiRand. Private release is useful only on the largest log.

Keywords: Causal recommendation, causal graph recommendation, doubly robust estimation, exposure bias, missing not at random, graph-based recommendation, exposure invariance.

1. Introduction

Graph-based recommender systems represent users and items as nodes and logged interactions as edges, and they score a user-item pair by aggregating signals along multi-hop walks (He et al., 2020; Shen et al., 2021). The logged graph, however, is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded, so the graph records the exposure policy as much as the user’s taste. A recent survey of causal learning in graph-based recommendation (Yu et al., 2026) traces several open problems of the field, among them fairness, interpretability and out-of-distribution behaviour, to message passing over this observational graph, and it lists theoretical guarantees and privacy as open directions.

Two families of corrections exist. *Loss-level* debiasing re-weights the training objective by inverse propensities or doubly robust (DR) terms (Schnabel et al., 2016; Wang et al., 2019b; Saito et al., 2020), but it leaves the propagation operator of a graph model untouched. *Propagation-level* debiasing changes the propagated graph itself, for example by weighting each edge by its inverse propensity (Kim et al., 2022) or by learning popularity-aware

aggregation weights (Zhou et al., 2023; Islam et al., 2026). This paper starts from a simple observation about the second family. An inverse-propensity edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , but a walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. The same happens for DR edge estimates. Walks that repeat an edge are part of every multi-hop operator: the back-and-forth step $u \rightarrow j \rightarrow u$ of a three-hop walk is one. Their excess is largest where the propensity is smallest, that is, on the rarely exposed edges that debiasing is meant to recover.

We make this observation precise and remove the resulting bias exactly. Four points qualify its practical reach, and we state them up front. First, the guarantees need nuisances (propensities, imputation, degree weights and score constants) that do not depend on the log. Cross-fitting the propensity and the imputation suffices if the degree weights are fixed a priori, but not if they are computed from the log, which our experiments do; in that case the correction no longer reduces bias in simulation. A sample split does supply independent nuisances, and we treat this variant, DRUP-split, as the estimator that the theory covers. Second, for ranking the *unexposed* candidates of a user with inverse-propensity edges, the repeated walks through the candidate vanish at three hops, and the bias only appears from five hops on; for DR edges it is present at three hops. Third, on four public benchmarks removing the bias does not change top- K accuracy beyond 0.005 nDCG, also under exposure caps, and the most accurate method differs by dataset. Fourth, the correction matters where score values are used, but only under oracle conditions: on a semi-synthetic log built from KuaiRec’s fully observed matrix, uncorrected propagation overestimates the full-exposure utility of an allo-

cation by 18–29%, and the corrected operator is within 2%, provided that the propensities are correct and the propagation weights are fixed a priori; with estimated propensities all operators overstate utility by 100–540%. The contribution is therefore a diagnosis and an exact correction at the level of scores, not an accuracy gain, and we report where the evidence for each further property is weak: the thinning audit of exposure invariance, the private release and the certificates (Section 7).

Contributions.

1. *Diagnosis.* We give the exact conditional bias of inverse-propensity and DR graph propagation on unexposed candidates. It is unbounded as propensities vanish, and under a propensity clip τ its worst case is $\Theta(1/\tau)$ (Theorem 5).
2. *Estimator.* DRUP (Doubly Robust, walk-Unbiased Propagation) uses $Y_e^2 = Y_e$ for binary outcomes and replaces every repeated edge by its idempotent unbiased counterpart. The three-hop operator has the asymptotic cost of the uncorrected one (one item Gram matrix plus $O(mn)$ reductions) and factorises into a public item operator and per-user corrections. Möbius inversion over the coincidence patterns of walk indices makes the correction exact for any odd number of hops (Theorem 3). A sample split supplies nuisances that do not depend on the edge estimates.
3. *Guarantees.* With nuisances that do not depend on the log, DRUP is edge-wise doubly robust for propagation over the full-exposure graph (Theorem 1); on unexposed candidates it is unbiased up to the imputation error at the candidate itself (Corollary 2); its bias under exposure

dependence within a user’s history is of second order in the imputation error (Proposition 4); we bound its variance and clipping bias (Theorem 6); and its expected scores do not respond to interventions on an item’s exposure (Theorem 7). Properties that follow from linearity alone (attributions, certificates, joint differential privacy, capped allocation) hold for any linear operator and are in the Supplementary Material.

4. *Evaluation.* We check the guarantees and their scope by Monte Carlo, including the gap between the theory and cross-fitted nuisances; test score values and score-based decisions on a semi-synthetic KuaiRec log with known propensities; and compare DRUP with up to 26 tuned baselines on Coat (Schnabel et al., 2016), Yahoo! R3 (Marlin and Zemel, 2009), KuaiRand (Gao et al., 2022b) and KuaiRec (Gao et al., 2022a), whose test data are random or fully observed, with per-split validation for every method, an analysis of tuning budgets, and user-level tests with Holm correction. Real-data thinning interventions, with a negative control, test exposure invariance.

2. Related work

We review the lines of work that DRUP touches. Table 1 summarises how DRUP relates to representative methods.

Graph-based recommendation. Graph neural recommenders learn user and item representations by propagating embeddings over the user–item graph. NGCF (Wang et al., 2019a) stacks non-linear propagation layers, and LightGCN (He et al., 2020) shows that removing feature transformations and

non-linearities improves accuracy, which reduces propagation to a weighted sum of powers of a normalised adjacency. UltraGCN (Mao et al., 2021) approximates infinite-layer propagation with a constraint loss, and SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022) add contrastive views. Closed-form models are strong competitors of trained propagation: EASE (Steck, 2019) learns an item–item matrix with a zero diagonal in closed form, and EDLAE (Steck, 2020) generalises the constraint by penalising the diagonal, which, like our diagonal correction of G , removes an item’s own contribution to its score; GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023) interpret collaborative filtering as graph signal filtering, and graph convolutions can trade accuracy for diversity (Isufi et al., 2021). Reproducibility studies show that carefully tuned simple baselines often match graph models (Anelli et al., 2023). All these models propagate over the *logged* graph. DRUP keeps the linear propagation form of LightGCN and of closed-form filters but propagates over an unbiased estimate of the full-exposure graph, and we compare it with EASE and GF-CF on both graphs.

Bias and debiasing in recommendation. Recommendation logs are missing not at random: users see what the system exposed and rate what they chose to rate (Marlin and Zemel, 2009). Surveys organise the resulting biases and their remedies (Chen et al., 2023) and cast debiasing as causal inference (Gao et al., 2024). Propensity-based methods re-weight the loss by inverse exposure probabilities (Schnabel et al., 2016), possibly self-normalised to reduce variance (Swaminathan and Joachims, 2015); Saito et al. (2020) extend them to implicit feedback, and Liang et al. (2016) model exposure as a latent variable. Doubly robust learning combines inverse propensities

with an imputation model (Wang et al., 2019b), and later variants reduce its bias and variance (Guo et al., 2021; Dai et al., 2022; Li et al., 2023) or learn information-theoretic representations (Wang et al., 2020). A small uniformly logged sample can supervise the debiasing itself (Chen et al., 2021). Causal approaches to popularity bias remove its direct effect by counterfactual inference (Wei et al., 2021), adjust for it as a confounder (Zhang et al., 2021), or disentangle interest from conformity (Zheng et al., 2021). In this journal, popularity debiasing has been studied from the user and item side jointly (Boratto et al., 2021), with a controllable accuracy trade-off (Okamura et al., 2024), through invariant learning with biased imputation (Bai et al., 2025), and by causal deconfounding of multi-criteria ratings (Guo et al., 2026). These methods correct the loss or the output of a model. DRUP applies IPS and DR estimation to the *edges* of the propagated graph, and its control-variate family interpolates between the two.

Debiasing inside graph propagation. Closest to our work are methods that change neighbour aggregation. NAVIP (Kim et al., 2022) weights each edge by its inverse propensity inside LightGCN. r-AdjNorm (Zhao et al., 2022) shows that the normalisation exponent of the adjacency controls how strongly popularity is amplified, and tunes it; graph convolutions provably amplify popularity bias (Chen et al., 2024), and deconfounding can reduce the amplification of user-level biases (Wang et al., 2021). APDA (Zhou et al., 2023) learns an adaptive inverse-popularity weight per edge, CAGED (Que et al., 2025) derives aggregation weights from a backdoor adjustment, and DPAA (Islam et al., 2026) learns interaction- and layer-wise weights to counter the amplification of popularity by repeated message passing. These

methods change *which* adjacency is propagated and target popularity-related metrics. None of them analyses what happens to an unbiased edge estimate when the re-weighted adjacency is multiplied with itself over several hops. Theorem 5 shows that the product is then biased, by a term of order $1/p$ that concentrates on rarely exposed items, and DRUP removes this term exactly for any odd number of hops. Our linear propagation baseline over the logged graph searches the normalisation exponent and is therefore a training-free r-AdjNorm; we also train r-AdjNorm and NAVIP.

Unbiased estimation of polynomials under missingness. The walk correction belongs to a classical idea: an unbiased estimator of a polynomial in unknown quantities must avoid products of the same noisy measurement, as in U-statistics (Hoeffding, 1948). For covariance matrices estimated from Bernoulli-masked data, the off-diagonal entries of the masked Gram matrix need a $1/p^2$ correction and the diagonal a $1/p$ correction (Lounici, 2014); our diagonal-corrected item operator G is the analogue for inverse-propensity and DR edges with heterogeneous propensities. Unbiased subgraph counts from sampled edges (Tsourakakis et al., 2009), and the passage from homomorphism counts to injective counts by Möbius inversion on the partition lattice (Rota, 1964), are the combinatorial counterparts of our K -hop construction. Removing walks that revisit vertices or edges is also standard in network analysis: non-backtracking operators (Krzakala et al., 2013) and self-avoiding-walk statistics (Hopkins and Steurer, 2017) discard backtracking walks to control the spectrum or the variance. Our correction differs in that it keeps the repeated walks and replaces each repeated factor by its unbiased counterpart, so that the estimator targets the full walk sum,

repeats included. In recommendation, Schnabel and Bennett (2020) debias item-to-item co-occurrence statistics with propensities and a small annotated sample. The reduction itself is therefore not new. What is new is its combination with inverse-propensity and DR edge estimates and degree weights in graph recommendation, the exact bias of the uncorrected operators, the public-operator form, and the consequences for exposure invariance. Our treatment of nuisances follows double machine learning, whose cross-fitting we adopt (Chernozhukov et al., 2018), and our clipping and control-variate choices mirror doubly robust off-policy evaluation (Dudík et al., 2011).

Fairness in recommendation. Fairness notions in recommendation cover users and items, groups and individuals, and outcomes and exposure (Wang et al., 2023); a unifying account of their measurement is given by Amigó et al. (2023), and provider fairness and its relation to data characteristics have been analysed in Gómez et al. (2022); Deldjoo et al. (2021). On the item side, exposure can be allocated in proportion to merit (Singh and Joachims, 2018), amortised over time (Biega et al., 2018) or under uncertainty about merit (Singh et al., 2021), and two-sided allocations can be guaranteed by constrained matching (Patro et al., 2020); expected exposure evaluates stochastic rankings against merit (Diaz et al., 2020), and multi-stakeholder views weigh users, providers and the platform (Abdollahpouri et al., 2020; Ekstrand et al., 2022). Fairness can also be audited by counterfactual reasoning (Cornacchia et al., 2023). DRUP contributes a causal item-side property of the merit *estimate*, exposure invariance: its expected scores do not respond to interventions on an item’s exposure (Theorem 7). It is not a fairness criterion. Merit-based allocation can be applied to any

merit estimate, and what it requires is an estimate whose errors do not depend on past exposure; exposure invariance is one such property, relative to the full-exposure target F^* as the notion of merit. It can increase the concentration of recommendations, and the distributive mechanism is an explicit allocation such as the exposure-capped b -matching of Theorem S5, related to (Patro et al., 2020).

Explanations, robustness and privacy. Explainable recommendation is surveyed by Zhang and Chen (2020). For graph models, GNNExplainer (Ying et al., 2019) and CF-GNNExplainer (Lucic et al., 2022) search for important or minimal sets of edges, PRINCE (Ghazimatin et al., 2020) finds minimal sets of a user’s actions whose removal changes a random-walk recommendation, and CountER (Tan et al., 2021) optimises counterfactual explanations of aspect-based models. Counterfactual explanations have a legal motivation (Wachter et al., 2018) and hidden assumptions (Barocas et al., 2020), and Shapley attributions depend on their baseline (Kumar et al., 2020). Recommenders can be manipulated by injected profiles (Lam and Riedl, 2004; Li et al., 2016; Fang et al., 2018), and shilling groups can be detected with graph convolutions (Wang et al., 2022); PORE (Jia et al., 2023) certifies robustness by ensembling models trained on sub-sampled data. Private recommenders perturb item–item statistics (McSherry and Mironov, 2009), factorisation models (Liu et al., 2024) or matrix completion under joint differential privacy (Jain et al., 2018), and federated approaches keep ratings on the device (Chai et al., 2021). Any linear propagation operator with fixed nuisances, DRUP included, admits exact removal attributions, closed-form certificates and a jointly private release of its item operator with an ana-

Table 1: DRUP and representative related methods. “loss”: the correction acts on the training objective; “output”: it acts on the predictions; “learned”: learned aggregation weights; “–”: not applicable. Properties that follow from linearity (exact attributions, certificates, joint DP) hold for every linear operator, including linear LightGCN, GF-CF and DR adjacency, and are not listed.

Method	debases propagation	unbiased beyond 1 hop	doubly robust	exposure invariance
LightGCN, NGCF, UltraGCN, SGL, SimGCL	no	no	no	no
linear LightGCN, EASE, GF-CF, BSPM	no	no	no	no
IPS, SNIPS, DR, MRDR, StableDR	loss	–	DR variants	no
MACR, PDA, DICE	output	–	no	no
NAVIP	edge weights	no	no	no
r-AdjNorm, APDA, CAGED, DPAA	learned	no	no	no
DR adjacency (uncorrected)	DR edges	no	edge-wise	no*
DRUP (ours)	DR edges	any odd K	yes [†]	in expectation [†]

*Over-corrects (Theorem 7); invariant in our real-data audits. [†]Under Assumptions 1–2 (e.g., the sample split) and unclipped propensities.

lytically calibrated Gaussian mechanism (Balle and Wang, 2018; Hsu et al., 2014); we state and test these properties in the Supplementary Material.

3. Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome $Y_e \in \{0, 1\}$ (whether u would like i if it were shown) and an observation indicator $O_e \in \{0, 1\}$. The log reveals O and $O \odot Y$. What O records depends on the platform: on KuaiRec it is an impression chosen by the system, on Coat and Yahoo! R3 it is a rating that the user chose to give,

so the “exposure” mechanism modelled by p is then user self-selection rather than the platform’s display policy. All statements below are relative to the mechanism that generates O ; we keep the word exposure for brevity.

Assumption 1 (Conditionally independent exposure). *Conditionally on Y , the exposures O_e are independent Bernoulli variables with propensities $p_e = \Pr(O_e = 1 \mid Y) > 0$.*

The propensity may depend on the pair’s own outcome, so the log may be missing not at random. All expectations below are over O given Y .

Assumption 2 (Fixed nuisances). *The propensity estimates $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$, the edge weights C and the constants a, b of the score below are fixed given Y , i.e., they do not depend on O (for instance, they are fitted on an independent log). We use $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees d taken from $\hat{Y}$.*

The weights C also enter the target below, so with degrees from $\hat{Y}$ the target is indexed by the imputation. A normalisation fixed a priori (a design choice, or degrees from an independent source) makes the target independent of $\hat{Y}$; Section 6.3 compares both.

Remark 1 (Scope of the guarantees and the experimental protocol). All guarantees below are stated under Assumption 2. Our main experimental protocol does not satisfy it, in four ways. (i) $\hat{p}$ and $\hat{Y}$ are cross-fitted over ten random folds of pairs: the nuisances used at a pair of fold k are fitted on the other nine folds (Chernozhukov et al., 2018). This makes the nuisance at a pair independent of that pair’s exposure, which is what a one-hop estimate

needs, but not of the exposures of the other edges of a walk, so a product of several edge estimates can be biased although every single edge estimate is unbiased. (ii) The degree weights sum the imputation over all pairs of a user or an item, so they depend on the exposures of all these pairs, including the pair at which they are used. (iii) One of the searched degree sources takes degrees from edge estimates. (iv) The constants a, b are computed from the scores (Section 4). In simulation, the resulting dependence bias is of the same order as the repeated-walk bias, and under this protocol with imputation-based degrees the corrected operator is not less biased than the uncorrected one (Section 6.2); a control in which only the degree weights are fixed a priori restores the correction under cross-fitting, so (ii) and (iii), more than (i), account for the failure. The theorems therefore describe DRUP with nuisances from an independent log, or from the sample split introduced below; for the cross-fitted protocol they give no guarantee, and our experiments with it evaluate accuracy and the empirical behaviour of the scores, not unbiasedness.

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph,

$$F^* = a\tilde{Y} + bT^*, \quad T^* = \tilde{Y}\tilde{Y}^\top\tilde{Y}, \quad (1)$$

with fixed constants $a, b \geq 0$. F^* is the linear LightGCN score one would compute if every item had been shown to every user. It is a normative choice: it treats potential outcomes as fixed and ignores preferences that exposure itself shapes (Chaney et al., 2018; Perdomo et al., 2020) (Section 7).

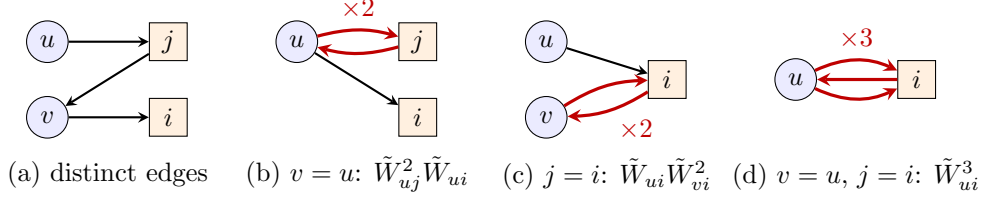


Figure 1: The four kinds of three-hop walks from user u to item i . Only walks of type (a) consist of distinct edges. In (b)–(d) an edge estimate is multiplied with itself (red), and its power has the wrong mean. DRUP replaces each repeated factor $\tilde{W}_e^k$ by $C_e^k W_e$.

4. DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e (Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Where the plug-in operator fails. The plug-in three-hop operator $(\tilde{W} \tilde{W}^\top \tilde{W})_{ui}$, with $\tilde{W} = C \odot W$, sums the products $\tilde{W}_{uj} \tilde{W}_{vj} \tilde{W}_{vi}$ over all walks $u-j-v-i$. If $v \neq u$ and $j \neq i$ the three edges are distinct and, by independence, the product is unbiased. The other walks repeat an edge (Figure 1): for $v = u$ the edge (u, j) is traversed twice, for $j = i$ the edge (v, i) , and for $v = u, j = i$ the edge (u, i) three times. A repeated factor $\tilde{W}_e^k$ has mean $C_e^k \mathbb{E}[W_e^k] \neq C_e^k Y_e$; for IPS, $\mathbb{E}[W_e^2] = Y_e/p_e$. Because $Y_e^k = Y_e$ for binary outcomes, the unbiased replacement of $\tilde{W}_e^k$ is simply $C_e^k W_e$.

Idempotent walk correction. Let $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected three-hop operator subtracts the excess of each walk type:

$$\hat{T} = \tilde{W} \tilde{W}^\top \tilde{W} - \tilde{W} \odot (r \mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1} \kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The second term removes the excess of type (b) walks, the third that of type (c), and the last that of type (d); the subtracted Δ inside the first two avoids counting type (d) twice. The DRUP score is $s = a\tilde{W} + b\hat{T}$. In the experiments $a = 1/c_1$ and $b = \beta/c_3$, where c_1 and c_3 are the mean absolute values of the one- and three-hop terms over all scored users and β is selected on validation data. The constants are global, so every user’s scores are the same linear combination of the two terms, and within-user rankings depend on them only through b/a ; a data-dependent c_3/c_1 thus acts as a data-dependent rescaling of the grid of β . Because c_1 and c_3 are computed from the log, however, $\mathbb{E}[\hat{T}/c_3] \neq T^*/c_3$ in general, and the score-level statements below hold for fixed a, b . Constants computed from the imputed graph $C \odot \hat{Y}$ satisfy Assumption 2 whenever $\hat{Y}$ does; Section 6.2 compares both choices.

Sample-split variant. A log rarely comes with an independent log for its nuisances. A sample split provides one at a cost. Draw a random set A of pairs, each pair with probability q , fit $\hat{Y}$ (and $\hat{p}$, where it is estimated) on the pairs in A only, replace the edge estimates of the pairs in A by $\hat{Y}$, take the degrees from $\hat{Y}$, and compute c_1 and c_3 on the imputed graph $C \odot \hat{Y}$. Conditionally on A and on the exposures in A , the remaining edge estimates are independent of every nuisance, so Assumption 2 holds and the results below apply, with $\xi_e = \hat{Y}_e - Y_e$ on the pairs in A . The price is a first-order imputation bias on a fraction q of the pairs and fewer exposures for the edge estimates. We call this variant DRUP-split and use $q = 0.2$. The guarantee holds for every fixed configuration; selecting the hyper-parameters on validation data, as every method in our comparison does, is outside it, as it is for any estimator.

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, \beta$

- 1: fit $\hat{p}$ (logistic exposure model) and $\hat{Y}$ (IPS-weighted additive ridge, optionally plus an IPS-weighted low-rank ALS residual), cross-fitted over ten folds of pairs
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$ $\triangleright$ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\zeta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\zeta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$ and lower-case rows belong to user u . Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1). Its off-diagonal entries are products of distinct edges; its diagonal is corrected from $\sum_u \tilde{W}_{ui}^2$ to $\sum_u C_{ui}^2 W_{ui}$, the analogue of the diagonal correction of a Gram matrix estimated from masked data (Lounici, 2014).

Control-variate family. More generally, $W_e^\lambda = O_e Y_e / \bar{p}_e + \lambda(\hat{Y}_e - O_e \hat{Y}_e / \bar{p}_e)$ is unbiased for every $\lambda \in [0, 1]$ under correct propensities. It interpolates between IPS ($\lambda = 0$) and DR ($\lambda = 1$, the only member that is also robust to propensity errors), and it equals the DR estimate with the shrunk imputation

$\lambda\hat{Y}$. With $\bar{p} = p$ its variance is $\frac{1-p}{p}(Y - \lambda\hat{Y})^2$. All results below hold for every member, and λ is selected on validation data.

Imputation. Theorem 1 holds for any fixed $\hat{Y}$, so the imputation is a design choice; its accuracy enters the variance and the clipping bias (Theorem 6). We use an additive model $\mu + b_u + b_i$ fitted by IPS-weighted ridge regression, or that model plus a rank-32 residual $UV^\top$ fitted by IPS-weighted alternating least squares. The additive imputation gives every user the same item order on unexposed candidates, which makes the one-hop DR term non-personalised; the low-rank imputation removes this limitation.

Complexity and scale. At three hops, one $n \times n$ Gram product and $O(mn)$ reductions, i.e., the cost of the uncorrected operator. The K -hop correction of Theorem 3 evaluates one tensor contraction per pair of equality patterns; we report its measured cost for $K = 5$ in Section 6.2. On KuaiRec (7,176 users, 10,728 items) one configuration takes a few seconds on four CPU cores once the nuisances are fitted. The dense W and G need $O(mn + n^2)$ memory, which limits the present implementation to catalogues of about 10^4 – 10^5 items. Because the correction only modifies diagonals and row and column sums, it carries over to sparse or low-rank representations of W , but we have not evaluated catalogues of 10^6 items.

5. Theory

Proofs are in Appendix A. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

5.1. Unbiasedness and its limits

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\xi_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every distinct edge contributing $Y_e + \xi_e$ once. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Recommendations rank the *unexposed* candidates of a user, and there the candidate’s own edge estimate is deterministic ($W_{ui} = \hat{Y}_{ui}$ for DR, 0 for IPS). The following corollary states what DRUP estimates on the candidate set.

Corollary 2 (Unexposed candidates). *Under the conditions of Theorem 1 with $\xi \equiv 0$,*

$$\mathbb{E}[s_{ui} \mid O_{ui} = 0] = F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) g_{ui},$$

$$g_{ui} = \frac{\partial F_{ui}^*}{\partial Y_{ui}} = C_{ui} \left(a + b \left(\sum_{v \neq u} C_{vi}^2 Y_{vi} + \sum_{j \neq i} C_{uj}^2 Y_{uj} + C_{ui}^2 \right) \right) \geq 0.$$

For IPS edges the same holds with $\hat{Y}_{ui} = 0$. The residual is the imputation error at the candidate itself. If p_e depends on Y only through Y_e , the log carries no information on Y_{ui} when $O_{ui} = 0$, so the residual is not identified and no estimator can remove it.

At three hops the walks through (u, i) are exactly those counted in g_{ui} . With IPS edges they vanish on unexposed candidates, so the uncorrected IPS operator has no repeated-walk bias there; from five hops on, walks repeat edges that are not incident to the candidate (for example $u-j-v-j-v-i$) and the bias reappears (Table 2).

Theorem 3 (Any odd number of hops). *For $K = 2r + 1$, a walk $U_0 - J_1 - U_1 - \dots - U_r - J_{r+1}$ has user variables $U_0, \dots, U_r$ and item variables $J_1, \dots, J_{r+1}$,*

with $U_0 = u$ and $J_{r+1} = i$. The equality pattern of an assignment is a pair π of set partitions of the user and item variables, and π fixes the multiplicity k_e of every distinct edge. Let $g_\pi(\sigma)$ be the unconstrained contraction, over the blocks of a coarser pattern $\sigma \geq \pi$, of $\prod_e C_e^{k_e} W_e$ (corrected) or of $\prod_e \tilde{W}_e^{k_e}$ (plug-in), the product running over the distinct edges of π . Then

$$\hat{T}^{(K)} = \tilde{W}(\tilde{W}^\top \tilde{W})^r + \sum_{\pi: \exists k_e > 1} \sum_{\sigma \geq \pi} \mu(\pi, \sigma) (g_\pi^{\text{corr}}(\sigma) - g_\pi^{\text{plug}}(\sigma)),$$

with μ the Möbius function of the product of the two partition lattices, satisfies $\mathbb{E}\hat{T}^{(K)} = (\tilde{Y}\tilde{Y}^\top)^r \tilde{Y}$ under the conditions of Theorem 1 with $\xi \equiv 0$.

The number of pairs (π, σ) is 5, 99 and 2,790 for $K = 3, 5, 7$, and for $K = 3$ the construction reduces to Eq. (3). Theorem 1, Corollary 2 and Theorem 7 carry over to any odd K .

Assumption 1 excludes slates of fixed size, where exposures in a user's history are negatively dependent (Coat, for instance, collects exactly 24 ratings per user). The next result bounds the resulting bias.

Proposition 4 (Exposure dependence within users). *Suppose that exposures are independent across users but may be dependent within a user's row, with $|\text{Cov}(O_e, O_f \mid Y)| \leq \rho p_e p_f$ for distinct e, f in the same row. With correct propensities,*

$$|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq \rho \sum_{j \neq i} C_{uj} \left(\sum_{v \neq u} C_{vj} C_{vi} \varepsilon_{vj} \varepsilon_{vi} + C_{uj} C_{ui} \varepsilon_{uj} \varepsilon_{ui} \right).$$

For slates of k items drawn uniformly without replacement, $\rho \leq 1/k$, and $\rho \leq 1$ for any negatively dependent design. For DR edges the bias is of second order in the imputation error; for IPS edges ($\varepsilon = Y$) it is of first order.

Theorem 5 (Repeated-walk bias of uncorrected propagation). *Let the propensities be correct and let (u, i) be a candidate with $O_{ui} = 0$. Relative to the conditional target of Corollary 2, the uncorrected DR operator $\tilde{W}\tilde{W}^\top\tilde{W}$ has conditional bias*

$$C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i) - C_{ui}^3\hat{Y}_{ui}(1 - \hat{Y}_{ui}^2),$$

$$K_i = \sum_{v \neq u} C_{vi}^2 V_{vi}, \quad R_u^{(i)} = \sum_{j \neq i} C_{uj}^2 V_{uj},$$

where $V_e = \text{Var}(W_e) = \frac{1-p_e}{p_e}\varepsilon_e^2$. Moreover: (a) $K_i \geq (1/\max_v p_{vi} - 1)\sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$, which is unbounded as the item's propensities vanish; (b) with clipped propensities $\bar{p} = \max(p, \tau)$, $\tau \leq 1/2$, the excess $C_e^2(\mathbb{E}[W_e^2] - \mathbb{E}[W_e])$ of a doubly traversed edge is at most $C_e^2(1/\tau - 1)\varepsilon_e^2$, with equality at $p_e = \tau$; the worst-case repeated-walk bias is therefore $\Theta(1/\tau)$; (c) for IPS edges the conditional bias vanishes at three hops. DRUP has zero bias relative to the same target.

The within-user ranking is affected through $C_{ui}\hat{Y}_{ui}K_i$, which up-weights items whose other raters were rarely exposed; $R_u^{(i)}$ depends on i only through one term.

Theorem 6 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/(4\tau^2)$, and $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$ if $\hat{p} = p$; (b) if $\hat{p} = p$, $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) if $\hat{p} = p$, so that clipping is the only source of bias, $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\xi}(1 + \bar{\xi})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\xi} = \max_e(1 - p_e/\tau)_+ \varepsilon_e$; (d) for any $\hat{p}$, $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2\exp(-2t^2/\sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

The variance scales with the imputation error ε^2 , not with Y^2 as for IPS, and τ trades variance against a clipping bias that vanishes when $\hat{Y}$ is accurate. The constants are worst-case and loose; Section 6.2 compares bound (b) with the Monte-Carlo variance.

Definition 1 (Exposure elasticity). For an intervention $do(p_i \leftarrow \pi p_i)$ on item i 's exposure mechanism, after which the estimator uses the post-intervention propensities ($\hat{p}_i \leftarrow \pi \hat{p}_i$) and otherwise the same nuisances, $\eta_i = \partial \log \mathbb{E}[s_{ui}] / \partial \log \pi$.

Theorem 7 (Exposure invariance). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, also conditionally on $O_{ui} = 0$: the expected score of every candidate is unchanged when item i 's exposure mechanism is scaled. Logged-graph propagation with fixed weights C has $\eta_i \geq 1$ (popularity amplification); normalising by logged degrees partly offsets it. Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui} K_i > 0$ (over-correction).*

Exposure invariance is a statement about expected scores. It is not a distributive fairness criterion, and it is conditional: when many propensities lie below the clip, $\pi p \geq \tau$ fails and invariance is lost (Section 6.6).

5.2. Structural properties

Given its nuisances, s is a fixed linear function of the edge estimates. Section S1 of the Supplementary Material states and proves the consequences, which hold for every linear propagation operator, corrected or not: exact attributions and minimal counterfactual explanations with frozen nuisances (Proposition S1, Theorem S2), conservative certificates against injected profiles (Theorem S3), joint differential privacy when the population-level nui-

sances, the hyper-parameters and the score constants are public (Theorem S4), and an exposure-capped allocation solved exactly by min-cost flow (Theorem S5). Because they are not specific to DRUP, we report their empirical behaviour in the Supplementary Material (Section S2) and use only the allocation in the main text, as a decision that compares scores across users (Section 6.3).

6. Experiments

6.1. Setup

Data. *Coat* (Schnabel et al., 2016) has 290 users, 300 items and 6,960 self-selected (MNAR) ratings, exactly 24 per user. The test data are 16 ratings per user on uniformly random items (MAR); we rank only the items the user did not rate, which leaves 14.7 candidates per user on average, and 225 users have at least one positive candidate. The dataset ships propensities, which we use. *Yahoo!R3* (Marlin and Zemel, 2009) has 311,704 self-selected ratings of 15,400 users on 1,000 songs (2.0% dense) and 54,000 ratings of 10 random songs for 5,400 of the users, in the version distributed with Chen et al. (2021). A random 5% of the MAR users is held out and used only to fit an outcome-dependent Naive-Bayes propensity (Schnabel et al., 2016); in a preliminary comparison on validation data it did not improve the propagation operators, and all reported results use the logistic exposure model. The remaining 5,130 test users have 10 candidates each, and 2,315 of them have a positive. *KuaiRec* (Gao et al., 2022a) provides the big matrix (7,176 users, 10,728 items, 10.3M logged views, 13.4% dense) as the MNAR log and a fully observed small matrix (1,411 users, 3,327 items) as the test set.

We rank all small-matrix items that the user never viewed in the log (3,314 candidates per user, 154 of them positive on average). *KuaiRand-Pure* (Gao et al., 2022b) is the only one of the four datasets whose log records platform exposure rather than self-selection: O is an impression of the standard recommendation feed and Y a click. We keep a random 10,000 of the users who also received randomly exposed videos, and all 7,583 videos; the standard log then has 511,863 impressions (0.68% dense, click rate 47%). The random impressions of the same period, restricted to videos the user was not shown in the standard log, are the test set (43.6 candidates per user, 17.5% clicked; 8,347 users have a positive). Positives are ratings ≥ 4 (Coat, Yahoo! R3), a watch ratio ≥ 2 (KuaiRec) and a click (KuaiRand). A random ranking reaches nDCG@5 of 0.306 on Coat and 0.352 on Yahoo! R3 and nDCG@20 of 0.046 on KuaiRec. The candidate sets of Coat and Yahoo! R3 are small, so their metric differences are compressed; KuaiRec ranks thousands of candidates. We report nDCG@5 on Coat and Yahoo! R3, nDCG@10 on KuaiRand and nDCG@20 on KuaiRec, and AUC and Recall in the Supplementary Material. On Coat and Yahoo! R3 the log is self-selected ratings: “exposure” there means that the user chose to rate the item, and exposure invariance is invariance to that choice, not to a platform’s allocation.

Protocol. The unbiased data are split into a validation part (30% of each user’s candidates on Coat, 30% of the test users otherwise) and a test part, over 10 (Coat) or 5 random splits. Every method selects its configuration on the validation part of each split, and trained models also select their epoch there: each split stops on its own validation score and never sees the validation or test users of another split. All fitted nuisances are cross-fitted over ten

folds of pairs, except for the sample-split variants, whose imputation (and propensity model, where it is estimated) is fitted on a random 20% of the pairs; only the latter satisfy the assumptions of the theory (Remark 1). We report the mean and standard deviation over splits of the test metric. Because paired tests over re-sampled splits are anti-conservative when the splits overlap (Nadeau and Bengio, 2003), differences are tested at the level of users, with a Holm correction over all comparisons of a dataset (Appendix B). These tests treat the training log, the fitted nuisances and the trained models as fixed: they account for the variation over test users, not for re-drawing the log or re-fitting the pipeline, which we assess only through the split and seed standard deviations. Progress claims in top- N recommendation have repeatedly failed against well-tuned simple baselines (Ferrari Dacrema et al., 2019; Rendle et al., 2019; Anelli et al., 2023); we therefore tune every method on the same validation data and report how many configurations each one searched.

Methods. We compare DRUP with up to 26 baselines (22 on KuaiRand, where eleven models were trained). *Non-personalised:* popularity and the imputation $\hat{Y}$ alone. *Trained with a pointwise loss:* MF, IPS-MF (Schnabel et al., 2016), DR-MF, DR-JL (Wang et al., 2019b), MRDR (Guo et al., 2021), iALS (Hu et al., 2008), LightGCN and DR-LightGCN (LightGCN with the DR loss). *Trained with a pairwise or contrastive loss:* MF, PDA (Zhang et al., 2021), MACR (Wei et al., 2021), LightGCN (He et al., 2020), r-AdjNorm (Zhao et al., 2022), NAVIP (Kim et al., 2022) and SimGCL (Yu et al., 2022), all with BPR. *Training-free on the logged graph:* linear LightGCN (the one- plus three-hop form of Section 4 on $O \odot Y$; with the searched

normalisation exponent it is a linear r-AdjNorm), EASE (Steck, 2019), GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023). *Training-free on a debiased graph*: IPS adjacency (NAVIP-style), IPS adjacency with the walk correction, EASE, GF-CF and BSPM on the DR graph W , the uncorrected DR adjacency, and the five-hop DR adjacency. Besides DRUP we report its five-hop version and DR adjacency and DRUP with the sample split of Section 4 (DRUP-split), over five draws of the split-off pairs. Grids, the number of configurations per method, and training details are in Appendix B. The training-free methods search between 1 and 6,300 configurations and the trained ones between 6 and 108. Trained models are evaluated after every epoch; each split keeps the best (configuration, epoch) on its own validation part and stops once its validation score has not improved for a fixed number of epochs.

6.2. Monte-Carlo checks

Table 2 uses a synthetic MNAR model with known propensities: 30 users, 40 items, low-rank preferences, and exposures driven by user activity, item popularity and the outcome itself (median propensity 0.06, a third of the pairs below 0.05). We draw 20,000 exposure matrices for three hops and 4,000 for five hops.

Over all entries, the uncorrected IPS and DR operators are biased by 2.3 and 1.8 times the mean target at three hops, and by 15 and 8 times at five hops. With the correction the residual bias (0.02–0.06) is at the level of the Monte-Carlo error. On unexposed candidates the picture is the one predicted by Corollary 2: the uncorrected IPS operator is unbiased at three hops (0.029, identical to IPS with the correction) but not at five hops (4.7),

Table 2: Monte-Carlo check of Theorems 1, 3, 5 and 7 and Corollary 2. Relative $|\text{bias}|$ is the mean absolute bias divided by the mean target, over all entries and conditionally on unexposed candidates (against the conditional target of Corollary 2). Kendall τ is the mean within-user rank agreement of one realisation with the full-exposure target on unexposed candidates. η is the exposure elasticity (separate model, 3,000 replications).

Estimator	rel. $ \text{bias} $, $K = 3$		rel. $ \text{bias} $, $K = 5$		Kendall τ	elasticity
	all	unexposed	all	unexposed	($K = 3$)	η
Logged graph	–	–	–	–	–	+1.777
IPS adjacency	2.335	0.029	14.641	4.667	0.024	+0.007
IPS + walk correction	0.028	0.029	0.060	0.059	0.024	+0.007
DR adjacency	1.756	1.137	8.350	5.053	0.199	-0.187
DRUP	0.019	0.019	0.039	0.040	0.122	+0.005

and the uncorrected DR operator is biased at both depths (1.1 and 5.1). The exposure elasticity is +1.78 for the logged graph, -0.19 for DR adjacency and +0.005 for DRUP. The K -hop engine reproduces the closed-form three-hop operator to 7×10^{-15} and brute-force enumeration of all five-hop walks to 10^{-13} . On a dense matrix of KuaiRec’s size (7,176 users, 10,728 items, 1,411 scored users, four CPU cores) the corrected three-hop operator takes 1.9s against 1.4s for the plug-in one, and the corrected five-hop operator 85s against 2.9s: the K -hop correction is exact but not free.

Unbiasedness does not buy ranking accuracy in this model. The rank agreement of a single realisation with the target is higher for the uncorrected DR operator (Kendall $\tau = 0.20$) than for DRUP (0.12), and at five hops 0.11 against 0.09. The reason is visible in Theorem 5: on unexposed candidates the bias is $C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i)$, which adds a positive multiple of the imputation to the score, and here the imputation is informative. The corrected operator

removes this term and, with it, some useful signal. For IPS edges, whose bias does not involve the imputation, the correction leaves the rank agreement unchanged (0.061 against 0.059 at five hops).

Theory against protocol. Table 3 measures how far the guarantees carry over to the way nuisances are fitted in practice. With nuisances from an independent log, the corrected operator’s bias is at the Monte-Carlo error (0.01–0.10) while the uncorrected operators are biased by 0.09–2.2 times the mean target. With the cross-fitted protocol of our main experiments this advantage disappears: the corrected operator is biased by 0.17–0.80 and the uncorrected one by 0.11–0.61, because the nuisances, and in particular the degree weights, depend on the exposures of the walk edges, and this dependence partly offsets the repeated-walk excess of the uncorrected operator. The dependence bias shrinks as users have more exposures (from 0.80 at 11 to 0.18 at 60 per user) but does not vanish. A control that fixes the degree weights a priori (Table S16 in the Supplementary Material) isolates the cause: with fixed weights, cross-fitting the propensity and the imputation alone preserves the correction (IPS with the correction 0.05 against 1.10 uncorrected; DRUP 0.05 against 0.89), whereas degrees taken from the cross-fitted $\hat{Y}$ or from the edge estimates destroy it (0.64–0.80 and 0.74–0.82). The failure of the main protocol is thus due to log-dependent degree weights, not to cross-fitting as such; fitting the nuisances in sample, without cross-fitting, breaks the correction even with fixed weights for DR edges (0.23). Degrees from edge estimates make the uncorrected operators much worse at low exposure (up to 4.1). The sample split restores the guarantee: relative to the target of Theorem 1 with $\hat{Y}$ on the imputed pairs, the bias of DRUP-split is 0.007–0.065, and

relative to the full-exposure target it is 0.07–0.18 with $q = 0.2$, against 0.07–1.09 for the uncorrected operator under the same split; $q = 0.5$ imputes too many pairs (0.13–0.42). The data-dependent constants matter as well: with independent nuisances and DR edges, the full score is biased by 0.18–0.35 when c_1, c_3 are computed from the scores and by 0.02–0.11 when they come from the imputed graph. On unexposed candidates, the conditional target of Corollary 2 differs from the full-exposure target by 12% (three hops) and 6% (five hops) of the mean target in this model, and it agrees with it in within-user order only up to Kendall $\tau = 0.63$ and 0.77: this is the part of F^* that no estimator can recover on the candidates actually ranked. Results with estimated propensities, which are misspecified in this model and bias every operator, are in Table S6 of the Supplementary Material.

Stress tests outside the assumptions of Theorem 1 are in the Supplementary Material (Table S5). With fixed-size slates, which make exposures within a user negatively dependent, IPS picks up a first-order bias (0.22) whereas DRUP stays at 0.024, as Proposition 4 predicts; with misspecified propensities DRUP is exactly unbiased when the imputation is exact and biased by 0.16 when it is noisy, against 0.57 for IPS and 1.9 for the uncorrected DR operator. The bounds of Theorem 6 were never violated but are loose by a factor of 70 to 8,000 (Table S7).

6.3. Score values on a semi-synthetic KuaiRec log

The real benchmarks cannot show whether score values are right, because F^* is unknown. We therefore use KuaiRec’s fully observed small matrix (1,411 users, 3,327 items, 4.6% positives) as the potential outcomes Y and draw exposure logs from a known MNAR mechanism fitted to the big matrix,

$\text{logit } p_{ui} = c + z_u + 1.5z_i + Y_{ui}$, with z the standardised logit exposure rates of the user and the item in the big log and c set for a density of 13% as in the big matrix (median propensity 0.064, all propensities at least $\tau = 0.02$). The target uses weights C fixed a priori (degrees of Y , $\alpha = 0.5$) and a, b fixed from the target. Each estimator uses either the same fixed C or degrees from its imputation (as in our real-data configurations), a rank-16 IPS-weighted imputation, and nuisances from an independent log, cross-fitted on the same log, or from the 20% sample split. Table 4 reports score errors and three uses of the scores.

Score values. With true propensities and fixed weights, the correction lowers the relative RMSE of the three-hop term by a factor of 4 to 12 (IPS 22.8 to 2.9, DR 28.2 to 6.3 with an independent log; DR 35.3 to 8.5 cross-fitted): the repeated factors of rarely exposed edges dominate the error as well as the bias. With 30 draws the remaining three-hop bias of the corrected operators is within Monte-Carlo error; at five hops the relative bias drops from 3.46 (DR) to 0.32 (DRUP). The difference is visible in the quantity a score-level user cares about. Summing the scores of a fixed allocation estimates its full-exposure utility, and the uncorrected operators overestimate it by 18–29% (random allocation) and 10–19% (popular items), whereas IPS with the correction and DRUP are within 2% of the truth. This holds with an independent log and, because C is fixed, also with cross-fitted nuisances (DRUP $-0.1 \pm 1.6\%$ against DR $+28.6 \pm 4.7\%$). The sample split removes the dependence but imputes a fifth of the pairs, and its imputation bias shifts the corrected estimates up by 38–47% (uncorrected: 53–64%).

What does not change. Two uses of the scores are unaffected. The exposure-capped allocation, which compares scores across users, loses 79–84% of the achievable full-exposure utility under every estimator: per-pair noise, not the repeated-walk bias, decides which pairs are allocated. Rankings do not improve either: with imputation-based weights, DR adjacency and DRUP reach nDCG@20 of 0.377 and 0.370. With weights taken from the imputation, finally, every operator estimates a different, imputation-indexed quantity; with an independent log or cross-fitted nuisances it is off by 40–57% in utility against the fixed-weight target, corrected or not, so score values need weights fixed a priori. With estimated propensities all estimators overstate utility by 100–540%, and the propensity model, not the walk correction, dominates the error (Table S12).

A sparser log. At a density of 3% (median propensity 0.008, 58% of the propensities below the clip $\tau = 0.01$) the conditions of the theorems fail, and clipping dominates (Table S13). The correction still lowers the three-hop RMSE by a factor of 7 to 32, but every utility estimate is biased: IPS with the correction by -28% , DR adjacency by $+331\%$ and DRUP by $+164\%$ with an independent log, in line with the clipping bias of Theorem 6(c).

6.4. Accuracy

Tables 5 and 6 report accuracy and the user-level tests; AUC and Recall are in the Supplementary Material and lead to the same conclusions.

The walk correction does not change accuracy. On every dataset DRUP and the uncorrected DR adjacency with the same grid are not significantly different (Coat $+0.0014$, KuaiRec -0.0001 , $p_{\text{Holm}} = 1$) and are equivalent

within ± 0.005 nDCG by two one-sided tests ($p_{\text{TOST}} < 10^{-3}$ on Yahoo!R3, KuaiRand and KuaiRec, 0.034 on Coat; Table S17), and on Yahoo!R3 and KuaiRand they coincide exactly, because validation selects the IPS end of the control-variate family ($\lambda = 0$) on every split and IPS edges have no three-hop repeated-walk bias on unexposed candidates (Corollary 2); on KuaiRand the selected clip is the smallest in the grid, 0.02, on all splits. The IPS adjacency and IPS with the correction give identical rankings on all datasets for the same reason. At five hops, where the correction is active for both edge types, the corrected operator is lower by 0.0003 nDCG@20 on KuaiRec ($p = 0.011$ before and $p_{\text{Holm}} = 0.16$ after the adjustment) and by 0.008 nDCG@5 on Coat ($p_{\text{Holm}} = 1$). The same holds under the sample split (Coat -0.003 , Yahoo!R3 -0.001 , KuaiRec -0.0001 ; all $p_{\text{Holm}} = 1$). Five hops do not add accuracy over three (KuaiRec 0.6376 against 0.6378).

The sample split. DRUP-split, the only variant covered by the guarantees, costs little on the larger and denser logs and much on the sparse one. On KuaiRec it reaches 0.6321 ± 0.0035 over five draws of the split-off pairs (0.627 to 0.636), 0.006 below the cross-fitted DRUP. For the first draw (0.6337) the gap is 0.0035 ($p_{\text{Holm}} = 0.003$), and that draw remains significantly above pointwise LightGCN ($+0.007$, $p_{\text{Holm}} < 10^{-3}$; Wilcoxon $p_{\text{Holm}} = 0.48$) and on par with DR-JL and MRDR ($p_{\text{Holm}} = 1$). On Coat its accuracy depends on the draw (0.544 ± 0.011 , from 0.527 to 0.558): with 24 logged ratings per user, which fifth of them is spent on the nuisances matters, and the first draw is not significantly different from the cross-fitted DRUP ($+0.009$, $p_{\text{Holm}} = 1$). On Yahoo!R3 it loses 0.062 (0.595 ± 0.004 ; $p_{\text{Holm}} < 10^{-3}$ for the first draw), because a fifth of the few logged exposures goes to the nuisances and the

imputation that replaces them is poor. Section 6.3 shows the corresponding cost for score values: the imputed pairs bias the utility estimates upwards.

Propagation over the DR graph is what helps on KuaiRec. On KuaiRec, whose test matrix is fully observed and whose candidate sets have 3,314 items, training-free propagation over the DR graph reaches nDCG@20 of 0.638 and is significantly better than every trained model at the user level: the best are the jointly learned doubly robust models DR-JL and MRDR (0.633; +0.004, $p_{\text{Holm}} = 0.003$; with the Nadeau–Bengio test over splits the margin over DR-JL is not significant, $p = 0.14$, and over MRDR $p = 0.006$, and it is smaller than their seed spread of 0.009 and 0.015), followed by point-wise LightGCN (0.627; +0.010, $p_{\text{Holm}} < 10^{-3}$). It is also better than every operator on the logged graph (GF-CF 0.614, BSPM 0.614, linear LightGCN 0.604) and than the imputation used alone (0.619). GF-CF on the same DR graph performs as well as DRUP (0.637, $p_{\text{Holm}} = 1$), so the gain comes from the DR graph and not from a particular filter. A control that propagates the imputation alone (every edge is $\hat{Y}$, no observed residual) reaches 0.6312: about two thirds of the 0.018 that DRUP gains over the imputation (0.6194) come from propagating the imputation, and one third from the observed residuals. On Coat the control equals the imputation (0.553 against 0.553; DRUP 0.547), and on Yahoo! R3 it is far below DRUP (0.528 against 0.657), where the IPS edges carry the signal.

KuaiRand. KuaiRand is the only dataset whose log records platform exposure. Its log is 0.7% dense, and as on Yahoo! R3 validation selects IPS edges ($\lambda = 0$) on every split, so DRUP, DR adjacency and IPS adjacency coincide (nDCG@10 0.4492). They are not significantly different from GF-CF

(0.4502) or BSPM (0.4498) on the logged graph, from iALS (0.4528) or from BPR-trained MF (0.4475; all $p_{\text{Holm}} = 1$), and they are significantly behind the BPR graph models: r-AdjNorm (0.4590; -0.010 , $p_{\text{Holm}} < 10^{-3}$), LightGCN (0.4569; -0.008) and NAVIP (0.4565; -0.008 ; both $p_{\text{Holm}} < 10^{-3}$). They are significantly ahead of popularity (0.4204), of the pointwise models (LightGCN 0.406, MF 0.371, DR-JL 0.376), of MACR (0.431) and of SimGCL (0.420). The sample split costs 0.009 (0.4400 ± 0.0007 over five draws; $p_{\text{Holm}} < 10^{-3}$), with DRUP-split and DR adjacency under the split again indistinguishable ($p_{\text{Holm}} = 1$). Only eleven baselines were trained on KuaiRand.

Coat and Yahoo!R3. On Coat (290 users, 15 candidates per user) the most accurate method is iALS (0.581), followed by BSPM on the logged graph, DR-LightGCN and the BPR graph models (0.566–0.573); DRUP reaches 0.547. With the Holm adjustment over all 124 comparisons, DRUP is significantly better than the pointwise MF, IPS-MF, DR-JL and MRDR and not significantly different from any other method, including iALS (-0.037 , $p = 0.033$ before adjustment) and GF-CF on the DR graph (-0.014 , $p = 0.001$ before and 0.11 after adjustment). On Yahoo!R3 the log is 2% dense and the imputation alone is below popularity (0.516 against 0.533), so almost every edge of a DR graph would be imputed from a poor model; validation therefore selects IPS edges ($\lambda = 0$) with cross-fitted W -degrees on every split, which lie outside Assumption 2. DRUP (0.657) is then on par with linear propagation over the logged graph (0.659), GF-CF (0.655) and iALS (0.663), and significantly behind the three graph models trained with BPR: r-AdjNorm (0.674; -0.019 , $p_{\text{Holm}} < 10^{-3}$), LightGCN (0.671; $p_{\text{Holm}} = 0.021$) and NAVIP

(0.670; $p_{\text{Holm}} = 0.029$). SimGCL (0.635) is below DRUP on this dataset ($p_{\text{Holm}} < 10^{-3}$).

Tuning budgets. The training-free operators search many more configurations than the trained models (Table B.12), because each configuration costs seconds rather than a training run. Table B.13 (Appendix B) gives the expected test score of DRUP when its configuration is selected from as many random configurations, or as many validation evaluations, as the most accurate trained model used. On Coat and Yahoo! R3 the larger grid does not help DRUP: with the budget of iALS (24 configurations) it reaches 0.549 on Coat, and with that of r-AdjNorm (72) 0.657 on Yahoo! R3, still below both. On KuaiRec the budget matters in configurations and not in evaluations. Counting a trained configuration with all its epochs as one, DRUP with the six configurations of DR-JL reaches only 0.618 against 0.633; counting every validation evaluation, as each epoch of a trained model is one, DRUP with the 103 evaluations of DR-JL reaches 0.636. On four CPU cores, fitting the nuisances takes about 50 minutes (14 for the cross-fitted propensity model and about 35 for the cross-fitted imputations of the grid, shared by all DR-type operators), after which a DRUP configuration costs about 3 s (21 minutes for all 384), against 130–210 s for training one LightGCN configuration.

Baselines. Three observations concern the baselines. First, the loss matters more than the architecture, and in opposite directions on the logs with few exposures per user and on KuaiRec: pairwise training is decisive on Yahoo! R3 (pointwise MF variants stay near 0.51, below popularity, whereas BPR-MF reaches 0.646 and BPR-trained LightGCN 0.671) and on KuaiRand (BPR-MF 0.448 against 0.371 pointwise, LightGCN 0.457 against 0.406),

whereas on KuaiRec the pointwise models beat their BPR versions (LightGCN 0.627 against 0.609, MF 0.622 against 0.596). Second, iALS is the strongest trained model on Coat and competitive on Yahoo! R3, but weak on KuaiRec (0.613), and the jointly learned DR models are the strongest trained models on KuaiRec but weak on the two small logs. No single trained baseline is strong on all datasets. Third, re-training the split-0 selection with other seeds shows how much of a trained model’s score is seed luck. On Yahoo! R3 the seeds lie within 0.013 of each other (0.007 for DR-JL and for the BPR graph models) and for the LightGCN variants on KuaiRec within 0.007; on Coat they spread by up to 0.05 for the graph models (r-AdjNorm 0.521–0.566), and on KuaiRec by 0.009 and 0.015 for DR-JL and MRDR, which learn their imputation jointly. The margin of DRUP over DR-JL and MRDR on KuaiRec (0.004) is therefore smaller than their seed variation, and differences on Coat should be read with this in mind; iALS and the training-free operators are deterministic given the log.

6.5. Does the correction matter on sparser logs?

Theorem 5 predicts a larger bias when propensities are small. We therefore thinned the KuaiRec log to 10% and 3% of its exposures, which lowers the median logged propensity from 0.48 to 0.045 and 0.013, re-fitted the nuisances and re-selected every operator (Table 7). The corrected and the uncorrected operators still rank almost identically: their top-20 lists overlap by at least 93%, the rank correlation is at least 0.999, and the accuracy differences stay below 0.0005 nDCG@20 in either direction, at three and at five hops. On these data the repeated-walk term is small compared with the score differences that decide the ranking, and the value of the correction lies

in the properties of the scores, not in the rankings they induce.

6.6. Exposure invariance on real data

Table 8 tests the prediction of Theorem 7 with a known intervention on exposure. The operators use the cross-fitted protocol, which the theorem does not cover (Remark 1), so this is an empirical check. The logged graph behaves as the theorem predicts for correlational propagation: with frozen nuisances $\eta = +1.00$ to $+1.03$ on all four datasets, and thinned items drop by 0.07–0.24 in mean within-user rank. On KuaiRec and Coat, DRUP is invariant to within a few hundredths ($\eta = -0.012$ and -0.008) and stays so when its nuisances are re-fitted on the thinned log, with the known or with re-estimated propensities where they are estimated ($|\eta| \leq 0.022$). This is what the theorem predicts, but the controls below show that the audit would give the same result for any estimator whose scores stay close to the imputation, so we read it as consistent with invariance, not as evidence for it. IPS adjacency is far from invariant ($\eta = +0.14$ on KuaiRec, $+0.90$ on Coat). The uncorrected DR adjacency is as invariant as DRUP on these data; its over-correction ($\eta = -0.19$ in Table 2) scales with $(1-p)/p \cdot \varepsilon^2$ (Theorem 5), and the selected clips and imputations keep it small here.

Yahoo!R3 and KuaiRand are the exceptions, and Table 10 explains the first. On Yahoo!R3, validation selects IPS edges ($\lambda = 0$) with a clip of 0.1–0.2, which binds for 82% of the logged pairs at $\tau = 0.2$. A clipped IPS edge is $O_e Y_e / \tau$, proportional to the exposure itself, so DRUP inherits the elasticity of the logged graph ($\eta = +0.96$). On Coat the clip binds just as often (77%), but with DR edges ($\lambda = 1$) and an accurate imputation the clipping bias is small (Theorem 6(c)) and invariance survives. Lowering the clip on

Yahoo! R3 reduces the elasticity to $+0.18 \pm 0.11$ at $\tau = 5 \cdot 10^{-4}$, where almost no propensity is clipped, at a cost of 0.044 nDCG@5. KuaiRand follows the same pattern without a clip sweep: validation selects IPS edges with the smallest clip in the grid (0.02), which binds for 57% of the treated logged pairs after the intervention, and DRUP is far from invariant ($\eta = +0.64$ with frozen nuisances, $+0.65$ when they are re-fitted; logged graph $+1.00$). The remaining elasticity and its large spread over the thinning draws come from heavy-tailed inverse-propensity weights on a log that is 2% dense. Exposure invariance is therefore a property of the estimator under its assumptions, and whether a tuned configuration has it must be checked; the thinning audit is a simple way to do so.

How informative is the audit? With frozen nuisances the audit updates the propensities to the known $p/2$, and an unbiased edge estimate is then unaffected by construction; a value near zero shows that the estimator is used as intended, not that the scores are right. Table 9 adds three controls. Popularity re-fitted on the thinned log has $\eta \approx 1$ and the imputation alone $\eta \approx 0$ on Coat and Yahoo! R3 (-0.16 on KuaiRec), which brackets the scale. The negative control, which keeps the propensities of the original log, should push a debiasing estimator towards $\eta = 1$. It does for IPS adjacency on all four datasets and for every operator on Yahoo! R3 and KuaiRand, whose selected configurations use IPS edges, but not for DR adjacency or DRUP on Coat ($\eta = -0.004$ and -0.008) or on KuaiRec (-0.087 and -0.088). There, the scores are almost a function of the imputation (rank correlation 0.98–0.99 with $\hat{Y}$), which is fitted on exposed pairs and reacts little to a halved exposure of some items; the audit cannot detect a misspecified propensity model

in this regime, and a near-zero η on Coat says little beyond the invariance of $\hat{Y}$. DRUP-split, whose edge estimates are less dominated by the imputation (rank correlation 0.76 on Coat, 0.58 on Yahoo!R3, 0.55 on KuaiRand), is invariant on Coat and KuaiRec ($|\eta| \leq 0.022$ frozen or re-fitted) and less elastic than the selected cross-fitted configuration on Yahoo!R3 and KuaiRand ($\eta = +0.32$ and $+0.33$ against $+0.96$ and $+0.64$). Both select IPS edges there, so the edge type does not explain the difference; a fifth of the split variant’s pairs carry the exposure-free imputation instead of a clipped IPS edge, which may account for it, but we did not isolate this. The audit is thus informative for IPS-dominated configurations and weak for configurations whose scores are close to the imputation, and it should be reported with these controls.

Table 11 shows the distributional side. On KuaiRec, the exposure-conditional bias drops from 0.83 (logged graph) and 0.80 (IPS) to 0.33 for DRUP and DR adjacency: given their true quality, items are ranked much less by how often they were shown. At the same time DRUP’s top-20 lists are concentrated on 1.2% of the candidate items, against 15% for the logged graph, because the estimated full-exposure scores are themselves concentrated on few items; on Coat its Gini coefficient is 0.60 against 0.34. Exposure invariance and exposure concentration are different properties, and DRUP improves the first at the expense of the second. The user-group gaps do not differ between operators. They are clearly non-zero on Coat (men 0.11 above women), on Yahoo!R3 (inactive users 0.046 below active ones) and on KuaiRand (inactive users 0.014 above active ones), under every operator, and not on KuaiRec. On KuaiRand the lists of DRUP, IPS

and the logged graph are almost equally concentrated (Gini 0.43, coverage 0.98), as the scores of all three are close to each other there.

7. Discussion

What the correction is for. On the four benchmarks the walk correction does not change top- K accuracy, and in simulation the uncorrected DR operator can even agree slightly better with the full-exposure ranking, because its bias adds weight to the imputation (Section 6.2). The correction can matter where the *value* of a propagated score is used rather than only its order within one user’s list, for instance when scores are aggregated into an estimate of full-exposure utility. It did not matter for the uses we tested that compare scores across users: the exposure-capped allocation loses 79–84% of the achievable utility under every estimator (Section 6.3), and the thinning audit does not separate the corrected from the uncorrected operator (Section 6.6). Where score values are used, the uncorrected estimate is biased by 1.8 to 15 times the target over all pairs in the model of Table 2 (0.09 to 2.2 times in the models of Table 3), and the bias grows with the number of hops and with small propensities. Two limits apply. On the unexposed candidates that are actually recommended, DRUP estimates the full-exposure score only up to the imputation error at the candidate (Corollary 2), which is 12% of the mean target at three hops in our simulation. And the guarantees assume nuisances that do not depend on the log: with nuisances cross-fitted on the same log, the corrected operator is not less biased than the uncorrected one in simulation (Section 6.2), so a user of the score values needs an independent log or the sample split of DRUP-split. We have no evidence

that the correction changes rankings, within users or under exposure caps: DRUP and DR adjacency are equivalent within ± 0.005 nDCG by two one-sided tests ($p < 10^{-3}$) on Yahoo!R3, KuaiRand and KuaiRec and marginally on Coat ($p = 0.034$; Table S17 in the Supplementary Material). A practitioner who only needs rankings loses nothing by using the uncorrected DR adjacency; one who aggregates score values or propagates over five hops or more needs the corrected operator for the estimate to be unbiased, with the caveats above. In both cases, most of the accuracy comes from training-free propagation over a doubly robust graph, whose gain over the imputation alone is two thirds from propagating the imputation.

Where the evidence is weak. Five results limit what the paper can claim, and we collect them here because each sits in a table far from the abstract. (i) The score-level benefit needs oracle conditions. In the semi-synthetic test the oracle target itself ranks the unexposed candidates perfectly (nDCG@20 of 1.0, because it contains the label), whereas the estimators reach 0.37–0.38; with degree weights taken from the imputation the corrected operator’s utility error is -49% against -44% uncorrected, and with estimated propensities it is $+100$ to $+540\%$ for all operators. (ii) The estimator the theory covers, DRUP-split, costs 0.062 nDCG on Yahoo!R3 and biases utility estimates upwards by 38–47% because a fifth of the pairs carry the imputation. (iii) The thinning audit has little power: the negative control, which keeps stale propensities, does not move the scores of DR adjacency and DRUP on Coat ($\eta = -0.008$) or KuaiRec ($\eta = -0.088$), their scores are correlated 0.98 (Coat) with the imputation, the uncorrected DR adjacency passes equally well, and the configurations selected on Yahoo!R3 and KuaiRand, the latter

the only log of platform impressions, are not invariant ($\eta = +0.96, +0.64$).

(iv) Private release is useful only on the largest log: it falls below popularity at every ϵ on Coat and Yahoo!R3 and for $\epsilon \leq 4$ on KuaiRec. The certificates cover 10–26% of users at 20 fake profiles, and at 200 fake profiles a DR or DRUP target reaches 39% of KuaiRec users, against 10% under IPS. (v) The statistical evidence is thinner than the user-level tests suggest. With the Nadeau–Bengio test over splits, DRUP is no longer significantly ahead of DR-JL on KuaiRec ($p = 0.14$ against $p_{\text{Holm}} = 0.003$; MRDR $p = 0.006$), and its 0.004 margin is smaller than the seed spread of DR-JL (0.009) and MRDR (0.015). We also note that DRUP searched 384 to 6,300 configurations against 6 to 108 for the trained models; at the six configurations of DR-JL its expected KuaiRec score is 0.618 against 0.633.

Stakeholders. Item providers. Exposure invariance means that an item’s expected score does not depend on how often it was observed; it is not equal exposure, and it comes with more concentrated lists (Table 11) that a re-trained system could entrench. Deployments should report concentration next to invariance and use an explicit allocation mechanism. *Users.* The removal attributions of the Supplementary Material describe the model with its nuisances frozen and are not an account of data deletion. *Platforms and auditors.* DRUP is cheap and training-free but needs correct propensities and nuisances independent of the log; the clip τ and the control-variate weight λ trade invariance against accuracy, and one option is to select the most accurate configuration subject to a bound on the elasticity measured in a thinning audit, whose limited power we described above. The audit is a first-party instrument: it needs the log, the propensities, the deployed hyper-parameters

and the ability to re-run the pipeline. Article 40 of the Digital Services Act (European Parliament and Council of the European Union, 2022) gives vetted researchers access to data of very large online platforms and search engines; whether it extends to re-running a proprietary pipeline is not settled by the text, and we do not claim that it does. The Act’s risk-assessment and audit obligations (Articles 34–37, Delegated Regulation (EU) 2024/436 (European Commission, 2024)) concern compliance of very large providers, not properties of estimators, and our results can at most inform a risk assessment.

Limitations. The analysis is static. It assumes a single logging policy and fixed potential outcomes, whereas recommenders are retrained on the feedback they generate (Chaney et al., 2018; Hazrati and Ricci, 2022) and can change the preferences they predict (Perdomo et al., 2020); extending the guarantees to adaptive and slate logging (Swaminathan et al., 2017) is open. Proposition 4 covers dependence within a user’s history, not dependence across users. The idempotent correction relies on binary outcomes; graded relevance would need separate unbiased estimates of the powers of Y . Items that were never exposed fall outside $\pi p \geq \tau$ and are scored by the imputation only. In our main protocol the nuisances are cross-fitted on the same log, the Yahoo!R3 configuration takes degrees from edge estimates, and the score constants are computed from the log, so the theorems do not cover these experiments (Remark 1); the sample split restores them at the price of an imputation bias on the split-off pairs and, on sparse logs, of accuracy. The significance tests are conditional on the single log and its fitted nuisances, and the methods searched grids of different sizes, which we analyse but cannot fully equalise (Section 6.4). The joint-DP guarantee requires pub-

lic nuisances, and on the small Coat data the public-nuisance variant loses much of its accuracy. The certificates are informative only against a handful of fake profiles (on KuaiRec, 95% of users at five profiles, 1% at fifty), and the variance and concentration bounds are loose by several orders of magnitude (Section 6.2). The target F^* is a linear LightGCN score of the full-exposure world, a normative choice that other stakeholders may not share. Like every method in our comparison, DRUP selects its hyper-parameters on a small unbiased sample; without such a sample, selection would need an off-policy estimate of the validation metric. The propensities shipped with Coat were estimated with the help of the uniform sample that we use for validation and testing (Schnabel et al., 2016), a possible leak that we cannot rule out. The within-user guarantee of Proposition 4 assumes known conditional propensities; for a fixed-size slate it is not a bound on the bias when the propensities are estimated. The target F^* contains the label of the pair it scores (the oracle ranks the unexposed candidates perfectly), so matching it on observed pairs is not the same as ranking well. Finally, our largest catalogue has about 10^4 items, and the dense implementation does not scale to catalogues of 10^5 or more.

Societal risks. Removing the dependence of expected scores on exposure does not make a recommender fair to item providers: DRUP’s lists are more concentrated than those of the logged graph (Gini 0.60 against 0.34 on Coat, 1.2% against 15% of the candidate items in the top-20 lists on KuaiRec), so a system retrained on its own output could entrench a small set of items. The user-group gaps we measure are the same under every operator. The privacy variant protects only the public item operator and is less accurate

than popularity on two of three datasets. We use only public, anonymised data sets and no human subjects.

8. Implications

For research. Propagation-level debiasing needs to consider the propagation operator itself, not only the edges it propagates: multiplying unbiased edge estimates does not give an unbiased multi-hop score. The repeated-walk bias is exactly removable, and the tools that remove it (idempotent reduction and Möbius inversion over walk-index coincidences) extend to polynomial filters of odd degree on binary outcomes built from inverse-propensity or DR edge estimates, as far as Theorem 3 goes. Our evaluation also shows that training-free propagation over a DR graph is a strong baseline against which trained debiased graph recommenders should be compared. Claims of exposure invariance can be probed on real data by thinning exposures with a known factor, but our controls show that such an audit has little power unless it includes a negative control and a configuration whose scores are not dominated by the imputation.

For practice. At three hops DRUP runs at the asymptotic cost of one item Gram matrix and needs no gradient training, so it can serve as an auditable baseline or as a candidate scorer. The choice between the corrected and the uncorrected operator depends on whether score values or only within-user orders matter; score values need nuisances that do not depend on the log, such as those of DRUP-split, since with nuisances cross-fitted on the same log they carry no unbiasedness guarantee. On sparse logs, exact exposure invariance requires small clips and costs accuracy (Table 10). The

structural properties (exact attributions, certificates, a private item operator and capped allocation) come with any linear propagation operator and hold for the model with its nuisances held fixed; platforms should report them together with that scope.

9. Conclusion

Inverse-propensity and doubly robust graph propagation are biased beyond one hop because walks that repeat an edge multiply its estimate with itself. We gave the exact bias, whose worst case grows as the inverse of the propensity clip, and removed it with an idempotent correction that is exact for any odd number of hops and, at three hops, has the asymptotic cost of the uncorrected operator. With nuisances that do not depend on the log, for instance from a sample split, the resulting estimator, DRUP, is edge-wise doubly robust for propagation over the full-exposure graph, robust to within-user exposure dependence to second order in the imputation error, and exposure-invariant in expectation; in simulation, cross-fitting the nuisances on the same log does not preserve these guarantees, so the main experiments, which cross-fit, test a heuristic, and the sample-split variant is the one the theory covers. On a semi-synthetic log with known exposure the correction removes the overestimate of full-exposure utility that uncorrected operators produce. On four benchmarks with unbiased test data the correction leaves top- K accuracy and rankings unchanged, and the most accurate method differs by dataset: iALS on Coat, BPR-trained graph models on Yahoo! R3 and KuaiRand, and training-free propagation over the DR graph on KuaiRec. Its contribution is at the level of scores. Future work includes

private nuisance models, certificates that remain informative for large catalogues, guarantees under adaptive logging, and combining the correction with learned propagation weights.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support code development and to polish the writing. All ideas and results are the author’s own. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- Abdollahpouri, H., Adomavicius, G., Burke, R., Guy, I., Jannach, D., Kamishima, T., Krasnodebski, J., Pizzato, L., 2020. Multistakeholder recommendation: Survey and research directions. *User Modeling and User-Adapted Interaction* 30, 127–158. doi:10.1007/s11257-019-09256-1.
- Amigó, E., Deldjoo, Y., Mizzaro, S., Bellogín, A., 2023. A unifying and general account of fairness measurement in recommender systems. *Information Processing & Management* 60, 103115. doi:10.1016/j.ipm.2022.103115.
- Anelli, V.W., Malitesta, D., Pomo, C., Bellogin, A., Di Sciascio, E., Di Noia, T., 2023. Challenging the myth of graph collaborative filtering: a reasoned and reproducibility-driven analysis, in: *Proceedings of*

- the 17th ACM Conference on Recommender Systems, ACM. pp. 350–361. doi:10.1145/3604915.3609489.
- Bai, T., Chen, W., Yang, C., Shi, C., 2025. Invariant debiasing learning for recommendation via biased imputation. *Information Processing & Management* 62, 104028. doi:10.1016/j.ipm.2024.104028.
- Balle, B., Wang, Y.X., 2018. Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising, in: *Proceedings of the 35th International Conference on Machine Learning*, pp. 394–403.
- Barocas, S., Selbst, A.D., Raghavan, M., 2020. The hidden assumptions behind counterfactual explanations and principal reasons, in: *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, ACM. pp. 80–89. doi:10.1145/3351095.3372830.
- Biega, A.J., Gummadi, K.P., Weikum, G., 2018. Equity of attention: Amortizing individual fairness in rankings, in: *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, ACM. pp. 405–414. doi:10.1145/3209978.3210063.
- Boratto, L., Fenu, G., Marras, M., 2021. Connecting user and item perspectives in popularity debiasing for collaborative recommendation. *Information Processing & Management* 58, 102387. doi:10.1016/j.ipm.2020.102387.
- Chai, D., Wang, L., Chen, K., Yang, Q., 2021. Secure federated matrix factorization. *IEEE Intelligent Systems* 36, 11–20. doi:10.1109/MIS.2020.3014880.

- Chaney, A.J.B., Stewart, B.M., Engelhardt, B.E., 2018. How algorithmic confounding in recommendation systems increases homogeneity and decreases utility, in: Proceedings of the 12th ACM Conference on Recommender Systems, ACM. pp. 224–232. doi:10.1145/3240323.3240370.
- Chen, J., Dong, H., Qiu, Y., He, X., Xin, X., Chen, L., Lin, G., Yang, K., 2021. AutoDebias: Learning to debias for recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 21–30. doi:10.1145/3404835.3462919.
- Chen, J., Dong, H., Wang, X., Feng, F., Wang, M., He, X., 2023. Bias and debias in recommender system: A survey and future directions. ACM Transactions on Information Systems 41, 1–39. doi:10.1145/3564284.
- Chen, J., Wu, J., Chen, J., Xin, X., Li, Y., He, X., 2024. How graph convolutions amplify popularity bias for recommendation? Frontiers of Computer Science 18. doi:10.1007/s11704-023-2655-2.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J., 2018. Double/debiased machine learning for treatment and structural parameters. The Econometrics Journal 21, C1–C68. doi:10.1111/ectj.12097.
- Choi, J., Hong, S., Park, N., Cho, S.B., 2023. Blurring-sharpening process models for collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1096–1106. doi:10.1145/3539618.3591645.

- Cornacchia, G., Anelli, V.W., Biancofiore, G.M., Narducci, F., Pomo, C., Ragone, A., Di Sciascio, E., 2023. Auditing fairness under unawareness through counterfactual reasoning. *Information Processing & Management* 60, 103224. doi:10.1016/j.ipm.2022.103224.
- Dai, Q., Li, H., Wu, P., Dong, Z., Zhou, X.H., Zhang, R., Zhang, R., Sun, J., 2022. A generalized doubly robust learning framework for debiasing post-click conversion rate prediction, in: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ACM. pp. 252–262. doi:10.1145/3534678.3539270.
- Deldjoo, Y., Bellogin, A., Di Noia, T., 2021. Explaining recommender systems fairness and accuracy through the lens of data characteristics. *Information Processing & Management* 58, 102662. doi:10.1016/j.ipm.2021.102662.
- Diaz, F., Mitra, B., Ekstrand, M.D., Biega, A.J., Carterette, B., 2020. Evaluating stochastic rankings with expected exposure, in: *Proceedings of the 29th ACM International Conference on Information & Knowledge Management (CIKM)*, pp. 275–284. doi:10.1145/3340531.3411962.
- Dodge, J., Gururangan, S., Card, D., Schwartz, R., Smith, N.A., 2019. Show your work: Improved reporting of experimental results, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2185–2194. doi:10.18653/v1/D19-1224.
- Dudík, M., Langford, J., Li, L., 2011. Doubly robust policy evaluation and

- learning, in: Proceedings of the 28th International Conference on Machine Learning, pp. 1097–1104.
- Efron, B., Stein, C., 1981. The jackknife estimate of variance. *The Annals of Statistics* 9, 586–596.
- Ekstrand, M.D., Das, A., Burke, R., Diaz, F., 2022. Fairness in information access systems. *Foundations and Trends in Information Retrieval* 16, 1–177. doi:10.1561/15000000079.
- European Commission, 2024. Commission delegated regulation (EU) 2024/436 of 20 october 2023 supplementing regulation (EU) 2022/2065 by laying down rules on the performance of audits for very large online platforms and very large online search engines. *Official Journal of the European Union*, L series.
- European Parliament and Council of the European Union, 2022. Regulation (EU) 2022/2065 on a single market for digital services (Digital Services Act). *Official Journal of the European Union*, L 277, 27 October 2022.
- Fang, M., Yang, G., Gong, N.Z., Liu, J., 2018. Poisoning attacks to graph-based recommender systems, in: Proceedings of the 34th Annual Computer Security Applications Conference, pp. 381–392. doi:10.1145/3274694.3274706.
- Ferrari Dacrema, M., Cremonesi, P., Jannach, D., 2019. Are we really making much progress? a worrying analysis of recent neural recommendation approaches, in: Proceedings of the 13th ACM Conference on Recommender Systems, ACM. pp. 101–109. doi:10.1145/3298689.3347058.

- Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., He, X., Mao, J., Chua, T.S., 2022a. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 540–550.
- Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P., He, X., 2022b. KuaiRand: An unbiased sequential recommendation dataset with randomly exposed videos, in: Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM), pp. 3953–3957. doi:10.1145/3511808.3557624.
- Gao, C., Zheng, Y., Wang, W., Feng, F., He, X., Li, Y., 2024. Causal inference in recommender systems: A survey and future directions. ACM Transactions on Information Systems 42, 1–32. doi:10.1145/3639048.
- Ghazimatin, A., Balalau, O., Saha Roy, R., Weikum, G., 2020. PRINCE: Provider-side interpretability with counterfactual explanations in recommender systems, in: Proceedings of the 13th International Conference on Web Search and Data Mining, pp. 196–204. doi:10.1145/3336191.3371824.
- Gómez, E., Boratto, L., Salamó, M., 2022. Provider fairness across continents in collaborative recommender systems. Information Processing & Management 59, 102719. doi:10.1016/j.ipm.2021.102719.
- Guo, S., Zou, L., Liu, Y., Ye, W., Cheng, S., Wang, S., Chen, H., Yin, D., Chang, Y., 2021. Enhanced doubly robust learning for debiasing post-click

- conversion rate estimation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 275–284. doi:10.1145/3404835.3462917.
- Guo, Z., Song, P., Feng, C., Yao, K., Liang, J., 2026. Causal inference for alleviating confounding bias in multi-criteria rating recommendation. *Information Processing & Management* 63, 104364. doi:10.1016/j.ipm.2025.104364.
- Hazrati, N., Ricci, F., 2022. Recommender systems effect on the evolution of users’ choices distribution. *Information Processing & Management* 59, 102766. doi:10.1016/j.ipm.2021.102766.
- He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M., 2020. LightGCN: Simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 639–648.
- Hoeffding, W., 1948. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics* 19, 293–325. doi:10.1214/aoms/1177730196.
- Hopkins, S.B., Steurer, D., 2017. Efficient Bayesian estimation from few samples: Community detection and related problems, in: 2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS), pp. 379–390. doi:10.1109/FOCS.2017.42.
- Hsu, J., Huang, Z., Roth, A., Roughgarden, T., Wu, Z.S., 2014. Private

- matchings and allocations, in: Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC), pp. 21–30.
- Hu, Y., Koren, Y., Volinsky, C., 2008. Collaborative filtering for implicit feedback datasets, in: 2008 Eighth IEEE International Conference on Data Mining (ICDM), pp. 263–272. doi:10.1109/ICDM.2008.22.
- Islam, M.A., Faruk, A.S., Medya, S., Zheleva, E., 2026. Debiasing message passing to mitigate popularity bias in GNN-based collaborative filtering. [arXiv:2605.11145](#).
- Isufi, E., Pocchiari, M., Hanjalic, A., 2021. Accuracy-diversity trade-off in recommender systems via graph convolutions. *Information Processing & Management* 58, 102459. doi:10.1016/j.ipm.2020.102459.
- Jain, P., Thakkar, O.D., Thakurta, A., 2018. Differentially private matrix completion revisited, in: Proceedings of the 35th International Conference on Machine Learning, pp. 2215–2224.
- Jia, J., Liu, Y., Hu, Y., Gong, N.Z., 2023. PORE: Provably robust recommender systems against data poisoning attacks, in: Proceedings of the 32nd USENIX Security Symposium, pp. 1703–1720.
- Kim, M., Oh, J., Do, J., Lee, S., 2022. Debiasing neighbor aggregation for graph neural network in recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 4128–4132. doi:10.1145/3511808.3557576.
- Krzakala, F., Moore, C., Mossel, E., Neeman, J., Sly, A., Zdeborová, L., Zhang, P., 2013. Spectral redemption in clustering sparse networks.

- Proceedings of the National Academy of Sciences 110, 20935–20940.
doi:10.1073/pnas.1312486110.
- Kumar, I.E., Venkatasubramanian, S., Scheidegger, C., Friedler, S., 2020. Problems with Shapley-value-based explanations as feature importance measures, in: Proceedings of the 37th International Conference on Machine Learning, pp. 5491–5500.
- Lam, S.K., Riedl, J., 2004. Shilling recommender systems for fun and profit, in: Proceedings of the 13th International Conference on World Wide Web (WWW), pp. 393–402.
- Li, B., Wang, Y., Singh, A., Vorobeychik, Y., 2016. Data poisoning attacks on factorization-based collaborative filtering, in: Advances in Neural Information Processing Systems.
- Li, H., Zheng, C., Wu, P., 2023. StableDR: Stabilized doubly robust learning for recommendation on data missing not at random, in: International Conference on Learning Representations (ICLR).
- Liang, D., Charlin, L., McInerney, J., Blei, D.M., 2016. Modeling user exposure in recommendation, in: Proceedings of the 25th International Conference on World Wide Web, pp. 951–961. doi:10.1145/2872427.2883090.
- Liu, H., Wang, Y., Zhang, Z., Deng, J., Chen, C., Zhang, L.Y., 2024. Matrix factorization recommender based on adaptive gaussian differential privacy for implicit feedback. Information Processing & Management 61, 103720. doi:10.1016/j.ipm.2024.103720.

- Lounici, K., 2014. High-dimensional covariance matrix estimation with missing observations. *Bernoulli* 20. doi:10.3150/12-bej487.
- Lucic, A., ter Hoeve, M.A., Tolomei, G., de Rijke, M., Silvestri, F., 2022. CF-GNNExplainer: Counterfactual explanations for graph neural networks, in: *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 4499–4511.
- Mao, K., Zhu, J., Xiao, X., Lu, B., Wang, Z., He, X., 2021. UltraGCN: Ultra simplification of graph convolutional networks for recommendation, in: *Proceedings of the 30th ACM International Conference on Information and Knowledge Management*, pp. 1253–1262. doi:10.1145/3459637.3482291.
- Marlin, B.M., Zemel, R.S., 2009. Collaborative prediction and ranking with non-random missing data, in: *Proceedings of the Third ACM Conference on Recommender Systems (RecSys)*, pp. 5–12.
- McDiarmid, C., 1989. On the method of bounded differences, in: *Surveys in Combinatorics*. Cambridge University Press, pp. 148–188.
- McSherry, F., Mironov, I., 2009. Differentially private recommender systems: Building privacy into the Netflix prize contenders, in: *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 627–636.
- Nadeau, C., Bengio, Y., 2003. Inference for the generalization error. *Machine Learning* 52, 239–281. doi:10.1023/a:1024068626366.
- Okamura, H., Maeda, K., Togo, R., Ogawa, T., Haseyama, M., 2024. Flexibly manipulating popularity bias for tackling trade-offs in recommendation.

- Information Processing & Management 61, 103606. doi:10.1016/j.ipm.2023.103606.
- Patro, G.K., Biswas, A., Ganguly, N., Gummadi, K.P., Chakraborty, A., 2020. FairRec: Two-sided fairness for personalized recommendations in two-sided platforms, in: Proceedings of The Web Conference 2020, pp. 1194–1204. doi:10.1145/3366423.3380196.
- Perdomo, J.C., Zrnic, T., Mendler-Dünner, C., Hardt, M., 2020. Performative prediction, in: Proceedings of the 37th International Conference on Machine Learning, pp. 7599–7609.
- Que, Y., Zhang, Y., Zhao, X., Ma, C., 2025. Causality-aware graph aggregation weight estimator for popularity debiasing in top-k recommendation, in: Proceedings of the 34th ACM International Conference on Information and Knowledge Management, ACM. pp. 2471–2481. doi:10.1145/3746252.3761155.
- Rendle, S., Zhang, L., Koren, Y., 2019. On the difficulty of evaluating baselines: A study on recommender systems. *arXiv:1905.01395*.
- Rota, G.C., 1964. On the foundations of combinatorial theory i. theory of möbius functions. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 2, 340–368. doi:10.1007/bf00531932.
- Saito, Y., Yaginuma, S., Nishino, Y., Sakata, H., Nakata, K., 2020. Unbiased recommender learning from missing-not-at-random implicit feedback, in: Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM), pp. 501–509.

- Schnabel, T., Bennett, P.N., 2020. Debiasing item-to-item recommendations with small annotated datasets, in: Fourteenth ACM Conference on Recommender Systems (RecSys), pp. 73–81. doi:10.1145/3383313.3412265.
- Schnabel, T., Swaminathan, A., Singh, A., Chandak, N., Joachims, T., 2016. Recommendations as treatments: Debiasing learning and evaluation, in: Proceedings of the 33rd International Conference on Machine Learning (ICML), pp. 1670–1679.
- Shen, Y., Wu, Y., Zhang, Y., Shan, C., Zhang, J., Letaief, K.B., Li, D., 2021. How powerful is graph convolution for recommendation?, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM), pp. 1619–1629.
- Singh, A., Joachims, T., 2018. Fairness of exposure in rankings, in: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 2219–2228. doi:10.1145/3219819.3220088.
- Singh, A., Kempe, D., Joachims, T., 2021. Fairness in ranking under uncertainty, in: Advances in Neural Information Processing Systems.
- Steck, H., 2019. Embarrassingly shallow autoencoders for sparse data, in: The World Wide Web Conference, ACM. pp. 3251–3257. doi:10.1145/3308558.3313710.
- Steck, H., 2020. Autoencoders that don’t overfit towards the identity, in: Advances in Neural Information Processing Systems 33 (NeurIPS), pp. 19598–19608.

- Swaminathan, A., Joachims, T., 2015. The self-normalized estimator for counterfactual learning, in: *Advances in Neural Information Processing Systems*.
- Swaminathan, A., Krishnamurthy, A., Agarwal, A., Dudík, M., Langford, J., Jose, D., Zitouni, I., 2017. Off-policy evaluation for slate recommendation, in: *Advances in Neural Information Processing Systems*.
- Tan, J., Xu, S., Ge, Y., Li, Y., Chen, X., Zhang, Y., 2021. Counterfactual explainable recommendation, in: *Proceedings of the 30th ACM International Conference on Information and Knowledge Management*, pp. 1784–1793. doi:10.1145/3459637.3482420.
- Tsourakakis, C.E., Kang, U., Miller, G.L., Faloutsos, C., 2009. Doulion: counting triangles in massive graphs with a coin, in: *Proceedings of the 15th ACM SIGKDD international conference on Knowledge discovery and data mining*, ACM. pp. 837–846. doi:10.1145/1557019.1557111.
- Wachter, S., Mittelstadt, B., Russell, C., 2018. Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology* 31, 841–887.
- Wang, S., Zhang, P., Wang, H., Yu, H., Zhang, F., 2022. Detecting shilling groups in online recommender systems based on graph convolutional network. *Information Processing & Management* 59, 103031. doi:10.1016/j.ipm.2022.103031.
- Wang, W., Feng, F., He, X., Wang, X., Chua, T.S., 2021. Deconfounded recommendation for alleviating bias amplification, in: *Proceedings of the*

- 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining, pp. 1717–1725. doi:10.1145/3447548.3467249.
- Wang, X., He, X., Wang, M., Feng, F., Chua, T.S., 2019a. Neural graph collaborative filtering, in: Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 165–174. doi:10.1145/3331184.3331267.
- Wang, X., Zhang, R., Sun, Y., Qi, J., 2019b. Doubly robust joint learning for recommendation on data missing not at random, in: Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 6638–6647.
- Wang, Y., Ma, W., Zhang, M., Liu, Y., Ma, S., 2023. A survey on the fairness of recommender systems. *ACM Transactions on Information Systems* 41, 52:1–52:43. doi:10.1145/3547333.
- Wang, Z., Chen, X., Wen, R., Huang, S.L., Kuruoglu, E.E., Zheng, Y., 2020. Information theoretic counterfactual learning from missing-not-at-random feedback, in: *Advances in Neural Information Processing Systems*.
- Wei, T., Feng, F., Chen, J., Wu, Z., Yi, J., He, X., 2021. Model-agnostic counterfactual reasoning for eliminating popularity bias in recommender system, in: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 1791–1800. doi:10.1145/3447548.3467289.
- Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., Xie, X., 2021. Self-supervised graph learning for recommendation, in: Proceedings of the

- 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 726–735. doi:10.1145/3404835.3462862.
- Ying, R., Bourgeois, D., You, J., Zitnik, M., Leskovec, J., 2019. GNNExplainer: Generating explanations for graph neural networks, in: Advances in Neural Information Processing Systems.
- Yu, D., Li, Q., Soundararaj, B., Pettit, C., Xu, G., 2026. A survey of causal learning in graph-based recommender systems. *ACM Computing Surveys* 58, 404:1–404:32. doi:10.1145/3840748.
- Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., Nguyen, Q.V.H., 2022. Are graph augmentations necessary?: Simple graph contrastive learning for recommendation, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1294–1303. doi:10.1145/3477495.3531937.
- Zhang, Y., Chen, X., 2020. Explainable recommendation: A survey and new perspectives. *Foundations and Trends in Information Retrieval* 14, 1–101.
- Zhang, Y., Feng, F., He, X., Wei, T., Song, C., Ling, G., Zhang, Y., 2021. Causal intervention for leveraging popularity bias in recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 11–20. doi:10.1145/3404835.3462875.
- Zhao, M., Wu, L., Liang, Y., Chen, L., Zhang, J., Deng, Q., Wang, K., Shen, X., Lv, T., Wu, R., 2022. Investigating accuracy-novelty performance for graph-based collaborative filtering, in: Proceedings of the 45th

International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 50–59. doi:10.1145/3477495.3532005.

Zheng, Y., Gao, C., Li, X., He, X., Li, Y., Jin, D., 2021. Disentangling user interest and conformity for recommendation with causal embedding, in: Proceedings of the Web Conference (WWW), pp. 2980–2991.

Zhou, H., Chen, H., Dong, J., Zha, D., Zhou, C., Huang, X., 2023. Adaptive popularity debiasing aggregator for graph collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 7–17. doi:10.1145/3539618.3591635.

Appendix A. Proofs

Throughout, expectations are conditional on Y , and “walk” means a three-hop walk $u - j - v - i$ unless stated otherwise. Its edges are (u, j) , (v, j) and (v, i) .

Equations (3) and (4). Split the plug-in sum $(\tilde{W}\tilde{W}^\top\tilde{W})_{ui} = \sum_{j,v} \tilde{W}_{uj}\tilde{W}_{vj}\tilde{W}_{vi}$ by walk type (Figure 1): (a) $v \neq u, j \neq i$; (b) $v = u, j \neq i$, giving $\sum_{j \neq i} \tilde{W}_{uj}^2 \tilde{W}_{ui}$; (c) $v \neq u, j = i$, giving $\sum_{v \neq u} \tilde{W}_{ui} \tilde{W}_{vi}^2$; (d) $v = u, j = i$, giving $\tilde{W}_{ui}^3$. The corrected estimator keeps (a) and replaces (b)–(d) by $\sum_{j \neq i} C_{uj}^2 W_{uj} \tilde{W}_{ui}$, $\sum_{v \neq u} \tilde{W}_{ui} C_{vi}^2 W_{vi}$ and $C_{ui}^3 W_{ui}$. The difference between plug-in and corrected sums is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui}) = \tilde{W}_{ui}(r_u - \Delta_{ui}) + \tilde{W}_{ui}(\kappa_i - \Delta_{ui}) + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$, which is Eq. (3). For Eq. (4), $(\tilde{w}_u G)_i = \sum_{j \neq i} \tilde{W}_{uj}(\tilde{W}^\top \tilde{W})_{ji} + \tilde{W}_{ui} \sum_v C_{vi}^2 W_{vi} =$

$(\tilde{W}\tilde{W}^\top\tilde{W})_{ui} - \tilde{W}_{ui}\kappa_i + \tilde{W}_{ui}\sum_v C_{vi}^2 W_{vi}$, and substituting into Eq. (3) cancels the κ_i terms. $\square$

Theorem 1.. By the decomposition above, the corrected estimator is a sum over walks of $\prod_e C_e^{k_e} W_e$, the product running over the *distinct* edges of the walk with multiplicities k_e . Under Assumptions 1–2 the W_e of distinct edges are independent, so the expectation factorises, with $\mathbb{E}W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e)/\bar{p}_e = Y_e + \xi_e$. The target walk monomial is $\prod_e C_e^{k_e} Y_e^{k_e} = \prod_e C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. The one-hop term is unbiased edge by edge. $\square$

Corollary 2.. Given $O_{ui} = 0$, $W_{ui} = \hat{Y}_{ui}$ is deterministic and the other edge estimates keep their distribution (Assumption 1). The edge (u, i) belongs to the one-hop term and to exactly the walks of types (b), (c) and (d). In each of them it contributes $C_{ui}^k \hat{Y}_{ui}$ instead of $C_{ui}^k Y_{ui}$, and every other distinct edge contributes its mean Y_e . After the idempotent reduction, F_{ui}^* is affine in Y_{ui} , so $\mathbb{E}[s_{ui} \mid O_{ui} = 0] = F_{ui}^*|_{Y_{ui} \leftarrow \hat{Y}_{ui}} = F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) \partial F_{ui}^* / \partial Y_{ui}$. The derivative collects aC_{ui} from the one-hop term, $bC_{ui} \sum_{j \neq i} C_{uj}^2 Y_{uj}$ from type (b), $bC_{ui} \sum_{v \neq u} C_{vi}^2 Y_{vi}$ from type (c) and bC_{ui}^3 from type (d). For IPS edges, $W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Theorem 3.. Fix u and i . For an assignment φ of values to the free variables, let $\pi(\varphi)$ be its equality pattern. Two walk edges (U_s, J_t) and $(U_{s'}, J_{t'})$ coincide if and only if $U_s, U_{s'}$ share a user block and $J_t, J_{t'}$ share an item block, so the distinct edges and their multiplicities k_e depend on $\pi(\varphi)$ only. For a pattern π , let $F_\pi(\varphi) = \prod_\epsilon X_{\varphi(\epsilon)}^{(k_\epsilon)}$, the product over the distinct edges ϵ of π , with $X^{(k)} = C^{\odot k} \odot W$ (corrected) or $X^{(k)} = \tilde{W}^{\odot k}$ (plug-in). The plug-in K -hop product equals $\sum_\pi S_\pi^{\text{plug}}$ and the corrected estimator is $\sum_\pi S_\pi^{\text{corr}}$, where $S_\pi =$

$\sum_{\varphi:\pi(\varphi)=\pi} F_{\pi}(\varphi)$. The two coincide for patterns without a repeated edge, which gives the form of $\hat{T}^{(K)}$ once each S_{π} is written through unconstrained sums. For $\sigma \geq \pi$ let $g_{\pi}(\sigma) = \sum_{\varphi:\pi(\varphi)\geq\sigma} F_{\pi}(\varphi)$: variables in the same block of σ are equal and values of different blocks are unrestricted, so $g_{\pi}(\sigma)$ is a single tensor contraction over the blocks of σ (with the blocks of U_0 and J_{r+1} fixed to u and i). Grouping the terms of $g_{\pi}(\sigma)$ by their exact pattern gives $g_{\pi}(\sigma) = \sum_{\tau\geq\sigma} f_{\pi}(\tau)$ with $f_{\pi}(\tau) = \sum_{\varphi:\pi(\varphi)=\tau} F_{\pi}(\varphi)$. Möbius inversion on the interval $[\pi, \hat{1}]$ of the product of the two partition lattices (Rota, 1964) yields $S_{\pi} = f_{\pi}(\pi) = \sum_{\sigma\geq\pi} \mu(\pi, \sigma) g_{\pi}(\sigma)$, where $\mu(\pi, \sigma) = \prod_B (-1)^{b_B-1} (b_B - 1)!$ over the blocks B of σ in both lattices and b_B is the number of blocks of π merged into B . Finally, for every φ with $\pi(\varphi) = \pi$ the distinct edges are independent, so $\mathbb{E}F_{\pi}^{\text{corr}}(\varphi) = \prod_{\epsilon} C_{\varphi(\epsilon)}^{k_{\epsilon}} Y_{\varphi(\epsilon)}$, which is the corresponding monomial of $(\tilde{Y}\tilde{Y}^{\top})^r \tilde{Y}$ since Y is binary. The number of contraction terms is the number of pairs (π, σ) with $\sigma \geq \pi$ and π containing a repeated edge, which is 5, 99 and 2,790 for $K = 3, 5, 7$. $\square$

Proposition 4. With correct propensities $\mathbb{E}W_e = Y_e$, and for distinct e, f in the same row $\text{Cov}(W_e, W_f) = (Y_e - \hat{Y}_e)(Y_f - \hat{Y}_f) \text{Cov}(O_e, O_f)/(p_e p_f)$, so $|\text{Cov}(W_e, W_f)| \leq \rho \varepsilon_e \varepsilon_f$. In a type (a) walk the edge (u, j) lies in row u and the edges $(v, j), (v, i)$ in row $v \neq u$; rows are independent, so $\mathbb{E}[W_{uj}W_{vj}W_{vi}] = Y_{uj}(Y_{vj}Y_{vi} + \text{Cov}(W_{vj}, W_{vi}))$ and the bias of the walk is at most $C_{uj}C_{vj}C_{vi}\rho\varepsilon_{vj}\varepsilon_{vi}$. A corrected type (b) walk contributes $C_{uj}^2 C_{ui} W_{uj} W_{ui}$ with two edges in row u , with bias at most $C_{uj}^2 C_{ui} \rho \varepsilon_{uj} \varepsilon_{ui}$. Type (c) walks have their two distinct edges in different rows and type (d) walks a single edge, so they are unbiased. Summing gives the bound. For k of n items drawn uniformly, $\Pr(O_e = O_f = 1) = k(k-1)/(n(n-1))$, so

$$|\text{Cov}(O_e, O_f)|/(p_e p_f) = (n - k)/(k(n - 1)) \leq 1/k. \quad \square$$

Theorem 5.. By the decomposition of Eq. (3), the plug-in minus the corrected operator is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$. Given $O_{ui} = 0$, $\tilde{W}_{ui} = C_{ui} \hat{Y}_{ui}$ is deterministic and the other Δ_e keep their distribution. With correct propensities and binary Y , $\mathbb{E}\Delta_e = C_e^2(\mathbb{E}W_e^2 - \mathbb{E}W_e) = C_e^2(\text{Var } W_e + Y_e^2 - Y_e) = C_e^2 V_e$ with $V_e = \frac{1-p_e}{p_e} \varepsilon_e^2$, which gives the stated bias, the last term being $C_{ui}^3(\hat{Y}_{ui}^3 - \hat{Y}_{ui})$. Part (a) uses $(1 - p)/p \geq 1/\max_v p_{vi} - 1$. For (b), with $\mu_e = \mathbb{E}W_e \in [0, 1]$, $\mathbb{E}W_e^2 - \mathbb{E}W_e = \text{Var } W_e - \mu_e(1 - \mu_e) \leq \text{Var } W_e = p_e(1 - p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is $\frac{1-p_e}{p_e} \varepsilon_e^2 \leq (1/\tau - 1)\varepsilon_e^2$; if $p_e < \tau \leq 1/2$ it is $p_e(1 - p_e)\varepsilon_e^2/\tau^2 < \tau(1 - \tau)\varepsilon_e^2/\tau^2$ because $p(1 - p)$ increases on $[0, 1/2]$. At $p_e = \tau$, $\mu_e = Y_e$ and the bound is attained; setting all $p_{vi} = \tau$ gives a bias $C_{ui} \hat{Y}_{ui}(1/\tau - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$, so the worst case is $\Theta(1/\tau)$. For (c), IPS gives $\tilde{W}_{ui} = 0$ and $C_{ui}^3 W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Theorem 6.. (a) $\text{Var}(W_e) = p_e(1 - p_e)\varepsilon_e^2/\bar{p}_e^2 \leq \varepsilon_e^2/(4\tau^2)$ for any $\hat{p}$, since $p(1 - p) \leq 1/4$ and $\bar{p}_e \geq \tau$. If $\hat{p} = p$, then $\bar{p} = \max(p, \tau)$: if $p_e \geq \tau$ the variance is at most $(1/\tau - 1)\varepsilon_e^2$, otherwise at most $p_e \varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. Without $\hat{p} = p$ the bound ε_e^2/τ can fail (e.g., $p = 0.5$, $\hat{p} = 0.01$, $\tau = 0.05$ gives $100\varepsilon_e^2 > 20\varepsilon_e^2$). (b) $\hat{T}_{ui}$ is multilinear in the independent W_e and hence affine in each of them. The Efron–Stein inequality (Efron and Stein, 1981) gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum over the walks through e of C -weights times at most two other independent W_f , and with $\hat{p} = p$, $\mu_f = Y_f \in [0, 1]$ and $\mathbb{E}W_f^2 = \mu_f^2 + \text{Var } W_f \leq 1 + 1/\tau$ by (a). Minkowski’s inequality bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum_{\text{walks}} |\kappa_{\text{walk}}| |\prod_f (Y_f + |\xi_f|) - \prod_f Y_f| \leq ((1 + \bar{\xi})^3 - 1) \sum_{j,v} C_{uj} C_{vj} C_{vi}$, and $(1 + x)^3 - 1 \leq 3x(1 + x)^2$. With $\hat{p} = p$, $|\xi_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing

O_e moves W_e by at most ε_e/τ . Every edge estimate satisfies $|W_f| \leq B$, so $|\partial_e \hat{T}| \leq B^2 \Lambda_e$, and McDiarmid’s inequality (McDiarmid, 1989) applies. $\square$

Theorem 7.. By Definition 1 the estimator uses the post-intervention propensities, and if $\pi p \geq \tau$ the clip does not bind, so they remain correct. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, and by Corollary 2, $\mathbb{E}[s \mid O_{ui} = 0]$ is F^* with Y_{ui} replaced by the fixed $\hat{Y}_{ui}$; neither depends on π . Logged graph with fixed C : $\mathbb{E}[O_e Y_e] = p_e Y_e$, and every walk that ends at i contains $k \geq 1$ edges incident to i , so its expectation scales as π^k and η_i is an average of such k . Uncorrected DR: by Theorem 5, $\mathbb{E}[s_{ui} \mid O_{ui} = 0]$ equals the DRUP target plus $C_{ui} \hat{Y}_{ui} (R_u^{(i)} + K_i(\pi))$ minus a term free of π ; $R_u^{(i)}$ involves no edge of item i , and $K_i(\pi) = \sum_{v \neq u} C_{vi}^2 \frac{1 - \pi p_{vi}}{\pi p_{vi}} \varepsilon_{vi}^2$ is strictly decreasing in π . $\square$

Appendix B. Experimental details

Training-free operators. All operators share the score $s = s_1/c_1 + \beta s_3/c_3$ (Section 4) with $\beta \in \{0, 0.03, 0.1, 0.3, 1, 3, 10\}$ or the three-hop term alone; the five-hop variants add $\gamma s_5/c_5$ with $\gamma \in \{0.03, 0.1, 0.3, 1, 3\}$. The grids are: normalisation exponent $\alpha \in \{0.3, 0.5, 0.7\}$ (Coat), $\{0.5, 0.7\}$ (Yahoo!R3, KuaiRand) and $\{0.3, 0.5\}$ (KuaiRec); clip $\tau \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ (Coat), $\{0.1, 0.2\}$ (Yahoo!R3), $\{0.02, 0.05, 0.1\}$ (KuaiRand) and $\{0.05, 0.1, 0.2\}$ (KuaiRec); imputation ridge $\lambda \in \{1, 5, 20\}$ (Coat), $\{20\}$ (Yahoo!R3, KuaiRand) and $\{5, 20\}$ (KuaiRec); additive or low-rank (rank 32) imputation; control-variate weight $\in \{0, 0.25, 0.5, 1\}$ on Yahoo!R3, $\{0, 0.5, 1\}$ on KuaiRand and 1 elsewhere; and degree source $\in \{\hat{Y}, \text{cross-fitted } W\}$ for the debiased operators. The degree source is a searched hyper-parameter. $\hat{Y}$ -degrees satisfy Assumption 2 whenever

$\hat{Y}$ does; cross-fitted W -degrees do not, since the weight at a pair then depends on the exposures of other pairs, so configurations with W -degrees (selected on Yahoo!R3) lie outside the theorems. Only $\hat{Y}$ -degrees make the certificates of Theorem S3 informative. EASE uses $\lambda \in \{10, 100, 1000\}$; GF-CF uses four settings of (ideal low-pass weight, singular vectors): $(0, 0)$, $(0.3, 64)$, $(0.3, 256)$ and $(1, 256)$. BSPM (Choi et al., 2023) uses blurring time $\in \{1, 2\}$, sharpening time $\in \{0, 0.1, 0.3\}$ and ideal low-pass weight $\in \{0, 0.3\}$ with one Euler step each. On the DR graph, EASE, GF-CF and BSPM search the same nuisance grid as DR adjacency. The sample-split variants are run for five draws of the split-off pairs; each draw is tuned separately, and the tables report the mean over draws. The propensity model is the one shipped with Coat and a logistic exposure model on user and item exposure rates elsewhere; on Yahoo!R3 it was selected on validation data against an outcome-dependent Naive-Bayes model. All nuisances are cross-fitted over ten random folds of pairs, except for the sample-split variants, whose imputation (and propensity model, where it is estimated) is fitted on a random 20% of the pairs.

Trained models. The grid combines the learning rates 10^{-2} , $3 \cdot 10^{-3}$ and 10^{-3} with the weight decays 10^{-5} , 10^{-4} and (Coat, and the MF models on Yahoo!R3) 10^{-3} . Embeddings have 64 dimensions (Coat: 32 or 64). The graph models with the widest search (LightGCN and r-AdjNorm on Yahoo!R3, the pointwise and BPR LightGCN on KuaiRec, all graph models on Coat) also search one to three layers and 64 or 128 dimensions (Coat: 32 or 64); the others use two layers. PDA searches $\gamma \in \{0.1, 0.2\}$, r-AdjNorm $r \in \{0.3, 0.7\}$, SimGCL the contrastive weight $\in \{0.1, 0.5\}$ (noise 0.1, tem-

perature 0.2, InfoNCE over at most 2,048 users and items of a batch) and MACR $c \in \{0.1, 0.3\}$; iALS searches the confidence $\alpha \in \{3, 10, 30\}$ and the ridge $\in \{1, 10, 100, 1000\}$. DR-JL and MRDR learn an MF imputation model jointly with the prediction model (Wang et al., 2019b; Guo et al., 2021). BPR models draw one uniformly sampled negative per positive. Batch sizes are 256 (Coat), 1,024 (Yahoo! R3), 8,192 (KuaiRand) and 32,768 (KuaiRec, with 2M sampled pairs per epoch for the pointwise losses); training runs for at most 100, 60, 40 and 30 epochs. Because the training log is the same for every split, each configuration is trained once and evaluated after every epoch on the validation part of every split. Each split stops on its own validation score (patience 5 on Coat and Yahoo! R3, 3 otherwise) and selects its own configuration and epoch; training continues until every split has stopped, and a split never uses epochs after its own stopping point, so no split’s selection depends on another split’s validation users, who may be its test users. The split-0 selection is re-trained with three (KuaiRec, KuaiRand: two) further seeds to measure seed variance. We did not run DICE, StableDR and methods that train on the uniform data (Cause, AutoDebias): in our protocol the uniform data serve only for selection and testing, for every method. Table B.12 lists the number of configurations each method was selected from.

Evaluation and tests. Coat splits each user’s candidates (30% validation), Yahoo! R3, KuaiRand and KuaiRec split the test users (30% validation), over 10, 5, 5 and 5 random splits. Users without a positive candidate are skipped. For a pair of methods, each test user’s metric difference is averaged over the splits in which the user is in the test part, and we report the mean difference over users with a 95% t -interval and the paired t -test. The Holm family is

every comparison run for a dataset, with all references together (DRUP, DR adjacency, five-hop DRUP and DRUP-split against every other method); the Wilcoxon signed-rank p -values are adjusted over the same family. Within-reference adjustments are in the result files. The Wilcoxon signed-rank test gives the same conclusions except for the five-hop DRUP against DR comparison on KuaiRec ($p = 0.66$ against 0.011) and for DRUP-split against pointwise LightGCN on KuaiRec ($p_{\text{Holm}} = 0.48$ against $3 \cdot 10^{-4}$). The corrected resampled t -test over split means (Nadeau and Bengio, 2003) is far more conservative with five or ten splits and is reported in the result files. All tests condition on the training log and the fitted nuisances.

Hardware and code. All experiments ran on four CPU cores with 16 GB of memory, except the KuaiRand debiased and sample-split operators, which ran on a Google Colab T4 GPU (same code and seeds; per-user ranking metrics are computed once per score matrix and match the per-user routine to 10^{-15}). GF-CF uses a randomised SVD, so its scores vary by about 0.002 between runs. Code, configurations and result files are available at the repository given on the title page.

Table 3: Monte-Carlo check of the nuisance protocol (Remark 1): relative $|\text{bias}|$ of the three-hop term against the full-exposure target with the same weights C , over all entries, with true propensities. Nuisances are fitted on an independent log (Assumption 2), cross-fitted over ten folds of pairs of the same log (the protocol of the main experiments), or fitted on a random 50% or 20% of the pairs of the same log (sample split). Model: users $\times$ items and mean exposures per user; the last model has three times larger propensities. The last two columns give the bias of the full score $s = s_1/c_1 + s_3/c_3$ with constants computed from the scores or from the imputed graph (2,000–3,000 replications).

Model	Nuisances	IPS edges		DR edges		$s, \text{data } c$	$s, \hat{Y} \ c$
		uncorr.	DRUP	uncorr.	DRUP		
40 $\times$ 60, 11	independent log	2.15	0.10	1.90	0.09	0.35	0.11
	cross-fitted (protocol)	0.39	0.64	0.61	0.80	0.62	0.88
	cross-fitted, W -degrees	3.12	0.81	4.11	0.73	0.49	2.69
	sample split, 50%	0.60	0.42	0.55	0.42	0.35	0.63
	sample split, 20%	1.09	0.17	0.90	0.17	0.29	0.24
60 $\times$ 120, 23	independent log	0.98	0.07	0.82	0.06	0.30	0.07
	cross-fitted (protocol)	0.31	0.52	0.42	0.61	0.42	0.58
	cross-fitted, W -degrees	1.51	0.69	1.80	0.58	0.40	1.09
	sample split, 50%	0.32	0.39	0.31	0.39	0.35	0.52
	sample split, 20%	0.54	0.18	0.47	0.18	0.26	0.26
60 $\times$ 120, 60	independent log	0.15	0.01	0.09	0.01	0.18	0.02
	cross-fitted (protocol)	0.11	0.17	0.12	0.18	0.20	0.12
	cross-fitted, W -degrees	0.17	0.18	0.11	0.12	0.18	0.08
	sample split, 50%	0.09	0.13	0.10	0.13	0.35	0.33
	sample split, 20%	0.09	0.07	0.07	0.07	0.23	0.14

Table 4: Semi-synthetic KuaiRec (30 draws of the log). RMSE T : root mean squared error of the three-hop term over all pairs, relative to the root mean square of the target (only for fixed C , where estimator and target use the same weights). Utility error: relative error (mean $\pm$ standard error over draws) of the estimated full-exposure utility $\sum_x s$ of a fixed random and a fixed popular allocation of 20 unexposed items per user. Regret: loss of full-exposure utility of the exposure-capped allocation computed from the estimated scores, relative to the one computed from F^* . nDCG@20 against Y on unexposed items. True propensities; estimated ones and a sparser log are in the Supplementary Material.

p	Nuisances	C	Estimator	RMSE T	utility error (%)			
					random	popular	regret	nDCG@20
true	independent log	fixed	IPS	22.8	+17.8±4.0	+10.3±0.4	0.795	0.097
			IPS+WC	2.9	+0.4±1.2	+0.4±0.3	0.795	0.097
			DR	28.2	+23.0±4.9	+17.8±0.5	0.791	0.097
			DRUP	6.3	-1.8±1.4	-0.0±0.4	0.795	0.107
true	independent log	$\hat{Y}$	IPS	–	-44.0±1.0	-41.3±0.2	–	0.120
			IPS+WC	–	-48.4±0.8	-44.4±0.2	–	0.120
			DR	–	-44.0±1.0	-40.4±0.2	–	0.377
			DRUP	–	-49.3±0.8	-44.6±0.2	–	0.370
true	cross-fitted	fixed	IPS	22.8	+17.8±4.0	+10.3±0.4	0.795	0.097
			IPS+WC	2.9	+0.4±1.2	+0.4±0.3	0.795	0.097
			DR	35.3	+28.6±4.7	+18.9±0.5	0.821	0.051
			DRUP	8.5	-0.1±1.6	+0.8±0.4	0.819	0.068
true	cross-fitted	$\hat{Y}$	IPS	–	-54.0±0.7	-46.5±0.1	–	0.191
			IPS+WC	–	-56.3±0.6	-48.5±0.1	–	0.191
			DR	–	-53.7±0.9	-45.4±0.1	–	0.320
			DRUP	–	-57.4±0.8	-48.8±0.1	–	0.322
true	sample split	fixed	IPS	20.6	+53.2±3.9	+55.9±0.6	0.822	0.093
			IPS+WC	2.5	+37.9±1.2	+47.2±0.6	0.821	0.091
			DR	103.6	+63.3±5.0	+63.7±0.8	0.840	0.030
			DRUP	8.8	+37.9±1.5	+47.0±0.6	0.837	0.048
true	sample split	$\hat{Y}$	IPS	–	+0.8±1.9	-10.4±0.6	–	0.154
			IPS+WC	–	-10.7±1.3	-17.1±0.5	–	0.150
			DR	–	+0.9±1.8	-10.3±0.6	–	0.274
			DRUP	–	-10.8±1.2	-17.2±0.5	–	0.272

Five hops (independent log, true p , fixed C): relative |bias| of T_5 , DR 3.46, DRUP 0.32.

Table 5: Accuracy on unbiased test data (mean $\pm$ std over splits; for the sample-split rows, mean $\pm$ std over five draws of the split-off pairs). Best in bold, second underlined. “–”: not run (five hops only where the three-hop DR operators are competitive). Every method is tuned per split on the same validation data; Table B.12 lists the number of configurations. Only the sample-split variants satisfy Assumption 2.

Method	Coat nDCG@5	Yahoo! R3 nDCG@5	KuaiRand nDCG@10	KuaiRec nDCG@20
<i>Non-personalised</i>				
Popularity	0.5367 $\pm$ 0.0095	0.5331 $\pm$ 0.0059	0.4204 $\pm$ 0.0019	0.5766 $\pm$ 0.0023
Imputation only	0.5532 $\pm$ 0.0097	0.5162 $\pm$ 0.0044	0.3637 $\pm$ 0.0015	0.6194 $\pm$ 0.0018
<i>Trained, pointwise loss</i>				
MF	0.4780 $\pm$ 0.0168	0.5087 $\pm$ 0.0075	0.3705 $\pm$ 0.0008	0.6221 $\pm$ 0.0036
IPS-MF	0.4713 $\pm$ 0.0160	0.5125 $\pm$ 0.0059	–	0.6252 $\pm$ 0.0018
DR-MF	0.5237 $\pm$ 0.0111	0.5093 $\pm$ 0.0042	0.3608 $\pm$ 0.0067	0.6219 $\pm$ 0.0025
DR-JL	0.4811 $\pm$ 0.0164	0.5225 $\pm$ 0.0043	0.3759 $\pm$ 0.0021	0.6332 $\pm$ 0.0026
MRDR	0.4869 $\pm$ 0.0187	0.5248 $\pm$ 0.0059	–	0.6328 $\pm$ 0.0020
iALS	0.5812$\pm$0.0101	0.6625 $\pm$ 0.0043	0.4528 $\pm$ 0.0017	0.6133 $\pm$ 0.0027
LightGCN	0.5389 $\pm$ 0.0107	0.5947 $\pm$ 0.0028	0.4061 $\pm$ 0.0014	0.6271 $\pm$ 0.0023
DR-LightGCN	0.5724 $\pm$ 0.0205	0.5528 $\pm$ 0.0091	–	0.6212 $\pm$ 0.0071
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.5174 $\pm$ 0.0206	0.6461 $\pm$ 0.0042	0.4475 $\pm$ 0.0027	0.5963 $\pm$ 0.0028
PDA	0.5181 $\pm$ 0.0138	0.6514 $\pm$ 0.0051	–	0.5815 $\pm$ 0.0027
MACR	0.5418 $\pm$ 0.0134	0.6221 $\pm$ 0.0038	0.4311 $\pm$ 0.0019	0.5816 $\pm$ 0.0033
LightGCN (BPR)	0.5683 $\pm$ 0.0132	<u>0.6708$\pm$0.0061</u>	<u>0.4569$\pm$0.0010</u>	0.6085 $\pm$ 0.0028
r-AdjNorm	0.5722 $\pm$ 0.0134	0.6742$\pm$0.0047	0.4590$\pm$0.0024	0.6083 $\pm$ 0.0031
NAVIP	0.5664 $\pm$ 0.0163	0.6701 $\pm$ 0.0039	0.4565 $\pm$ 0.0017	0.6186 $\pm$ 0.0024
SimGCL	0.5528 $\pm$ 0.0141	0.6348 $\pm$ 0.0048	0.4196 $\pm$ 0.0022	0.5637 $\pm$ 0.0031
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.5571 $\pm$ 0.0100	0.6585 $\pm$ 0.0058	0.4468 $\pm$ 0.0018	0.6036 $\pm$ 0.0028
EASE	0.5401 $\pm$ 0.0138	0.6478 $\pm$ 0.0061	0.4350 $\pm$ 0.0019	0.5213 $\pm$ 0.0032
GF-CF	0.5513 $\pm$ 0.0123	0.6552 $\pm$ 0.0039	0.4502 $\pm$ 0.0016	0.6141 $\pm$ 0.0025
BSPM	<u>0.5727$\pm$0.0090</u>	0.6507 $\pm$ 0.0047	0.4498 $\pm$ 0.0025	0.6143 $\pm$ 0.0026
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6132 $\pm$ 0.0030
IPS + walk correction	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6132 $\pm$ 0.0030
propagation over $\hat{Y}$ only (no residual)	0.5533 $\pm$ 0.0171	0.5280 $\pm$ 0.0043	–	0.6312 $\pm$ 0.0016
EASE on DR graph	0.5564 $\pm$ 0.0141	0.6072 $\pm$ 0.0033	0.4030 $\pm$ 0.0011	0.6025 $\pm$ 0.0022
GF-CF on DR graph	0.5608 $\pm$ 0.0083	0.6312 $\pm$ 0.0051	0.4468 $\pm$ 0.0012	0.6372 $\pm$ 0.0020
BSPM on DR graph	0.5624 $\pm$ 0.0112	0.6262 $\pm$ 0.0071	0.4483 $\pm$ 0.0016	0.6354 $\pm$ 0.0016
DR adjacency	0.5459 $\pm$ 0.0244	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	<u>0.6378$\pm$0.0020</u>
DRUP	0.5467 $\pm$ 0.0235	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6378 $\pm$ 0.0019
DR adjacency, 5 hops	0.5524 $\pm$ 0.0245	–	–	0.6379$\pm$0.0011
DRUP, 5 hops	0.5451 $\pm$ 0.0204	–	–	0.6376 $\pm$ 0.0011
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.5447 $\pm$ 0.0129	0.5952 $\pm$ 0.0037	0.4400 $\pm$ 0.0007	0.6322 $\pm$ 0.0035
DRUP-split	0.5443 $\pm$ 0.0112	0.5947 $\pm$ 0.0041	0.4400 $\pm$ 0.0007	0.6321 $\pm$ 0.0035

Table 6: User-level paired comparisons: mean difference of the test metric over users with a 95% interval, and the t -test p -value, Holm-adjusted over all comparisons run for the dataset (DRUP, DR adjacency and DRUP-split, and on Coat and KuaiRec also five-hop DRUP, each against every other method: 124 on Coat, 87 on Yahoo! R3, 75 on KuaiRand, 120 on KuaiRec). Users with a positive candidate who are in the test part of at least one split (225, 2,310, 8,321 and 1,410); sample-split rows use the first draw. Selected rows; all comparisons are in the result files.

Comparison	Coat		Yahoo! R3		KuaiRand		KuaiRec	
	Δ [95% CI]	P_{Holm}	Δ [95% CI]	P_{Holm}	Δ [95% CI]	P_{Holm}	Δ [95% CI]	P_{Holm}
DRUP vs DR adjacency (the correction)	+0.0014 [-0.002, +0.005]	1.000	+0.0000 [+0.000, +0.000]	1.000	+0.0000 [+0.000, +0.000]	1.000	-0.0001 [-0.000, +0.000]	1.000
DRUP vs DR adjacency, both 5 hops	-0.0081 [-0.015, -0.001]	1.000	-	-	-	-	-0.0003 [-0.001, -0.000]	0.170
DRUP-split vs DR adjacency, sample split	-0.0033 [-0.010, +0.003]	1.000	-0.0012 [-0.004, +0.001]	1.000	+0.0002 [-0.000, +0.001]	1.000	-0.0001 [-0.000, +0.000]	1.000
DRUP-split vs DRUP (cost of the split)	+0.0087 [-0.014, +0.032]	1.000	-0.0659 [-0.077, -0.055]	$< 10^{-3}$	-0.0092 [-0.012, -0.006]	$< 10^{-3}$	-0.0035 [-0.005, -0.002]	0.004
DRUP vs linear LightGCN, logged graph	-0.0123 [-0.051, +0.027]	1.000	-0.0022 [-0.005, +0.001]	1.000	+0.0025 [+0.001, +0.005]	0.267	+0.0346 [+0.030, +0.039]	$< 10^{-3}$
DRUP vs imputation only	-0.0082 [-0.017, +0.001]	1.000	+0.1411 [+0.125, +0.157]	$< 10^{-3}$	+0.0861 [+0.081, +0.091]	$< 10^{-3}$	+0.0179 [+0.016, +0.020]	$< 10^{-3}$
DRUP vs GF-CF	-0.0070 [-0.046, +0.032]	1.000	+0.0005 [-0.005, +0.006]	1.000	-0.0013 [-0.004, +0.001]	1.000	+0.0243 [+0.021, +0.028]	$< 10^{-3}$
DRUP vs BSPM	-0.0295 [-0.065, +0.006]	1.000	+0.0051 [-0.001, +0.011]	1.000	-0.0005 [-0.003, +0.002]	1.000	+0.0241 [+0.021, +0.027]	$< 10^{-3}$
DRUP vs GF-CF on DR graph	-0.0143 [-0.023, -0.006]	0.106	+0.0247 [+0.018, +0.032]	$< 10^{-3}$	+0.0025 [+0.001, +0.004]	0.043	+0.0004 [-0.001, +0.002]	1.000
DRUP vs iALS	-0.0372 [-0.071, -0.003]	1.000	-0.0065 [-0.014, +0.001]	1.000	-0.0031 [-0.006, +0.000]	1.000	+0.0249 [+0.021, +0.028]	$< 10^{-3}$
DRUP vs DR-JL	+0.0658 [+0.039, +0.093]	$< 10^{-3}$	+0.1350 [+0.120, +0.150]	$< 10^{-3}$	+0.0739 [+0.069, +0.079]	$< 10^{-3}$	+0.0044 [+0.002, +0.007]	0.003
DRUP vs LightGCN (pointwise)	+0.0073 [-0.014, +0.028]	1.000	+0.0616 [+0.048, +0.075]	$< 10^{-3}$	+0.0443 [+0.040, +0.049]	$< 10^{-3}$	+0.0103 [+0.007, +0.013]	$< 10^{-3}$
DRUP vs LightGCN (BPR)	-0.0252 [-0.058, +0.008]	1.000	-0.0146 [-0.023, -0.006]	0.021	-0.0081 [-0.011, -0.005]	$< 10^{-3}$	+0.0298 [+0.026, +0.033]	$< 10^{-3}$
DRUP vs r-AdjNorm	-0.0290 [-0.062, +0.004]	1.000	-0.0187 [-0.027, -0.010]	$< 10^{-3}$	-0.0101 [-0.013, -0.007]	$< 10^{-3}$	+0.0296 [+0.026, +0.033]	$< 10^{-3}$
DRUP vs NAVIP	-0.0236 [-0.056, +0.009]	1.000	-0.0138 [-0.022, -0.005]	0.029	-0.0077 [-0.011, -0.004]	$< 10^{-3}$	+0.0194 [+0.017, +0.022]	$< 10^{-3}$
DRUP vs SimGCL	-0.0077 [-0.033, +0.018]	1.000	+0.0219 [+0.012, +0.032]	$< 10^{-3}$	+0.0302 [+0.026, +0.034]	$< 10^{-3}$	+0.0742 [+0.070, +0.079]	$< 10^{-3}$
DRUP-split vs iALS	-0.0285 [-0.062, +0.005]	1.000	-0.0724 [-0.084, -0.061]	$< 10^{-3}$	-0.0123 [-0.016, -0.008]	$< 10^{-3}$	+0.0214 [+0.018, +0.025]	$< 10^{-3}$
DRUP-split vs DR-JL	+0.0745 [+0.043, +0.106]	$< 10^{-3}$	+0.0691 [+0.057, +0.081]	$< 10^{-3}$	+0.0647 [+0.060, +0.069]	$< 10^{-3}$	+0.0009 [-0.002, +0.003]	1.000
DRUP-split vs LightGCN (pointwise)	+0.0160 [-0.010, +0.042]	1.000	-0.0043 [-0.019, +0.010]	1.000	+0.0351 [+0.031, +0.040]	$< 10^{-3}$	+0.0068 [+0.004, +0.010]	$< 10^{-3}$
DRUP-split vs r-AdjNorm	-0.0203 [-0.052, +0.012]	1.000	-0.0846 [-0.096, -0.073]	$< 10^{-3}$	-0.0193 [-0.023, -0.016]	$< 10^{-3}$	+0.0261 [+0.023, +0.030]	$< 10^{-3}$

Table 7: KuaiRec with the log thinned to a fraction q of its exposures (nuisances re-fitted, additive imputation, $\tau \in \{0.05, 0.01\}$, $\lambda \in \{0, 1\}$, selected on validation). nDCG@20 of the uncorrected and corrected operators, and the lowest top-20 overlap and Spearman correlation between the two rankings at matched configurations.

thinning q	density	DR	DRUP	DR 5 hops	DRUP 5 hops	top-20 overlap	Spearman
1	13.38%	0.6302	0.6302	0.6306	0.6306	0.971	0.999
0.1	1.34%	0.6097	0.6102	0.6109	0.6111	0.935	0.999
0.03	0.40%	0.6073	0.6069	0.6085	0.6081	0.985	1.000

Table 8: Real-data intervention $do(p \leftarrow p/2)$: the logged exposures of a random half of the items are thinned with probability 1/2 (three to five draws). Rank shift: change of the treated items’ mean within-user rank relative to control items (0: invariant). η : elasticity of the mean score of treated relative to control candidates (1: exposure-proportional, 0: invariant). Frozen: imputation and degree weights C are kept, the propensities are updated to the known $p/2$; re-fitted: all nuisances are re-fitted on the thinned log, with the known propensity change or with propensities re-estimated on it. One configuration per operator (the one selected most often), all test data.

Data	Operator	frozen nuisances		re-fitted, known $p/2$		re-fitted, re-estimated $\hat{p}$	
		rank shift	η	rank shift	η	rank shift	η
Coat	logged graph	-0.0670	+1.031	-0.0585	+0.748	–	–
	IPS adjacency	-0.0652	+0.899	-0.0651	+0.900	–	–
	DR adjacency	+0.0021	-0.005	+0.0001	-0.019	–	–
	DRUP	+0.0036	-0.008	+0.0015	-0.022	–	–
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 94%						
Yahoo! R3	logged graph	-0.1611	+1.002	-0.0943	+0.508	-0.0943	+0.508
	IPS adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DR adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DRUP	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 95%						
KuaiRand	logged graph	-0.0826	+0.997	-0.0679	+0.711	-0.0679	+0.711
	IPS adjacency	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	DR adjacency	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	DRUP	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 57%						
KuaiRec	logged graph	-0.2408	+0.997	-0.1339	+0.500	-0.1339	+0.500
	IPS adjacency	-0.0462	+0.137	-0.0466	+0.141	-0.0657	+0.218
	DR adjacency	+0.0061	-0.013	+0.0001	-0.003	+0.0017	-0.006
	DRUP	+0.0057	-0.012	-0.0004	-0.002	+0.0011	-0.005
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 16%						

Table 9: Controls for the thinning audit (same intervention as Table 8, η averaged over draws). Negative control: the thinned log is scored with the propensities of the original log, so that the estimator is misspecified by a factor of two on treated pairs. Bottom: Spearman correlation of the scores with the imputation $\hat{Y}$ over all candidate pairs.

η	Coat	Yahoo! R3	KuaiRand	KuaiRec
popularity (re-fitted)	+0.993	+0.998	+1.000	+1.000
imputation only (re-fitted)	-0.010	-0.007	-0.014	-0.160
IPS adjacency, propensities not updated	+1.019	+1.002	+0.997	+0.997
DR adjacency, propensities not updated	-0.004	+1.002	+0.997	-0.087
DRUP, propensities not updated	-0.008	+1.002	+0.997	-0.088
DRUP-split, $\hat{Y}$ and C frozen	+0.022	+0.319	+0.328	-0.003
DRUP-split, re-fitted (known change)	+0.018	+0.319	+0.325	-0.007
DRUP-split, re-fitted (estimated)	–	+0.327	+0.331	-0.013
rank corr. with $\hat{Y}$: logged graph	0.30	0.27	0.36	0.36
rank corr. with $\hat{Y}$: DR adjacency	0.99	0.27	0.37	0.55
rank corr. with $\hat{Y}$: DRUP	0.98	0.27	0.37	0.54
rank corr. with $\hat{Y}$: DRUP-split	0.76	0.58	0.55	0.42

Table 10: Clip τ against exposure invariance for DRUP under the same intervention, with frozen nuisances (η : mean $\pm$ std over draws).

Data	Operator, τ	nDCG	rank shift	η	logged pairs clipped
Coat	logged graph	0.4917	-0.0646	+1.034 $\pm$ 0.043	–
	DRUP, $\tau = 0.01$	0.4191	+0.0071	-0.021 $\pm$ 0.008	1%
	DRUP, $\tau = 0.02$	0.4357	+0.0187	-0.054 $\pm$ 0.013	6%
	DRUP, $\tau = 0.05$	0.4599	+0.0104	-0.033 $\pm$ 0.008	12%
	DRUP, $\tau = 0.1$	0.4805	+0.0068	-0.021 $\pm$ 0.004	25%
	DRUP, $\tau = 0.2$	0.4891	+0.0046	-0.009 $\pm$ 0.002	77%
Yahoo! R3	logged graph	0.6590	-0.1604	+1.000 $\pm$ 0.011	–
	DRUP, $\tau = 0.0005$	0.6136	-0.0477	+0.181 $\pm$ 0.113	0%
	DRUP, $\tau = 0.002$	0.6140	-0.0519	+0.195 $\pm$ 0.099	1%
	DRUP, $\tau = 0.01$	0.6239	-0.0878	+0.508 $\pm$ 0.055	15%
	DRUP, $\tau = 0.05$	0.6500	-0.1274	+0.825 $\pm$ 0.017	51%
	DRUP, $\tau = 0.1$	0.6570	-0.1394	+0.901 $\pm$ 0.013	67%
	DRUP, $\tau = 0.2$	0.6572	-0.1494	+0.959 $\pm$ 0.009	82%
KuaiRec	logged graph	0.6026	-0.2412	+0.997 $\pm$ 0.003	–
	DRUP, $\tau = 0.01$	0.5309	-0.0000	-0.002 $\pm$ 0.002	2%
	DRUP, $\tau = 0.05$	0.6353	+0.0033	-0.007 $\pm$ 0.001	7%
	DRUP, $\tau = 0.1$	0.6382	+0.0059	-0.012 $\pm$ 0.000	10%
	DRUP, $\tau = 0.2$	0.6359	+0.0099	-0.021 $\pm$ 0.000	16%

Table 11: Exposure concentration and user groups. PRU: popularity–rank correlation. ECB (KuaiRec, where item quality is known from the fully observed matrix): partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. User gap: nDCG of group 0 minus group 1 (men minus women on Coat, inactive minus active users elsewhere) with a bootstrap 95% interval.

Data	Operator	nDCG	PRU	ECB	Gini@ K	user gap [95% CI]	coverage
Coat	logged graph	0.4917	0.394	–	0.340	+0.112 [+0.033, +0.190]	0.937
	IPS adjacency	0.4951	0.415	–	0.346	+0.108 [+0.027, +0.192]	0.937
	DR adjacency	0.4890	0.689	–	0.599	+0.111 [+0.024, +0.189]	0.667
	DRUP	0.4891	0.688	–	0.600	+0.111 [+0.025, +0.189]	0.663
Yahoo!R3	logged graph	0.6590	0.534	–	0.281	-0.046 [-0.071, -0.019]	1.000
	IPS adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DR adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DRUP	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
KuaiRand	logged graph	0.4475	0.776	–	0.430	+0.015 [+0.004, +0.026]	0.977
	IPS adjacency	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
	DR adjacency	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
	DRUP	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
KuaiRec	logged graph	0.6026	0.845	0.830	0.991	+0.008 [-0.012, +0.029]	0.152
	IPS adjacency	0.6123	0.875	0.801	0.992	+0.014 [-0.007, +0.035]	0.081
	DR adjacency	0.6385	0.802	0.329	0.994	+0.012 [-0.009, +0.031]	0.012
	DRUP	0.6382	0.803	0.331	0.994	+0.012 [-0.009, +0.031]	0.012

Table B.12: Number of configurations searched per method (per split).

Method	Coat	Yahoo! R3	KuaiRand	KuaiRec
Popularity (non-personalised)	1	1	1	1
Imputation only (non-personalised)	6	2	2	4
MF (trained, pointwise)	18	9	6	6
IPS-MF (trained, pointwise)	18	9	–	6
DR-MF (trained, pointwise)	18	9	6	6
DR-JL (trained, pointwise)	18	9	6	6
MRDR (trained, pointwise)	18	9	–	6
iALS (trained, pointwise)	24	12	12	12
LightGCN (trained, pointwise)	54	6	6	36
DR-LightGCN (trained, pointwise)	54	6	–	6
MF (BPR) (trained, pairwise)	18	9	6	6
PDA (trained, pairwise)	36	18	–	12
MACR (trained, pairwise)	36	18	12	12
LightGCN (BPR) (trained, pairwise)	54	36	6	36
r-AdjNorm (trained, pairwise)	108	72	12	12
NAVIP (trained, pairwise)	54	36	6	6
SimGCL (trained, pairwise)	108	12	12	12
linear LightGCN / r-AdjNorm (training-free)	24	16	16	16
EASE (training-free)	3	3	3	3
GF-CF (training-free)	4	4	4	4
BSPM (training-free)	12	12	12	12
IPS adjacency (NAVIP-style) (training-free)	720	64	96	192
IPS + walk correction (training-free)	720	64	96	192
propagation over $\hat{Y}$ only (no residual) (training-free)	1440	128	–	384
EASE on DR graph (training-free)	90	48	54	36
GF-CF on DR graph (training-free)	120	64	72	48
BSPM on DR graph (training-free)	360	192	216	144
DR adjacency (training-free)	1440	512	576	384
DRUP (training-free)	1440	512	576	384
DR adjacency, 5 hops (training-free)	6300	–	–	420
DRUP, 5 hops (training-free)	6300	–	–	420
DR adjacency, sample split (training-free)	720	256	288	192
DRUP-split (training-free)	720	256	288	192

Table B.13: Tuning budget. Expected test score of DRUP when its configuration is selected on validation data among k configurations drawn at random from its grid (exact, averaged over splits; Dodge et al. (Dodge et al., 2019)), with k the budget of the most accurate trained model. Configurations: a trained configuration keeps its best epoch per split, and every value of β is a separate DRUP configuration. Evaluations: every (configuration, epoch) of a trained model counts as one validation evaluation.

Data	best trained model	configurations		evaluations			
		k	DRUP	k	DRUP	DRUP, all	trained
Coat	iALS	24	0.5487	295	0.5451	0.5467	0.5812
Yahoo! R3	r-AdjNorm	72	0.6565	1765	0.6567	0.6567	0.6742
KuaiRec	DR-JL	6	0.6182	103	0.6362	0.6378	0.6332